

Look Ahead! Practical CCA-secure Steganography: Cover-Source Switching meets Lattice Gaussian Sampling

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Abstract. Steganography studies methods to not only protect the confidentiality of messages but also to conceal the very act of message transmission. Prior provably secure stegosystems are predominantly constructed based on a rejection sampling technique which achieves an encoding rate inversely proportional to the min-entropy of the cover channel. Furthermore, while replayable chosen-coverttext attack (RCCA) secure stegosystems for general channels can be constructed based on standard cryptographic assumptions, it is known [Berndt and Liśkiewicz, EUROCRYPT’18] that achieving (standard) CCA-security for channels with memory in the so-called non-look-ahead model is in general impossible and the only known CCA-secure construction crucially relies on the channels being memoryless.

In this work, we show that the impossibility on CCA-secure stegosystems can be circumvented, in the random oracle model, by dropping the non-look-ahead restriction and by restricting to a natural class of channels which we call “partially sampleable channels”. These capture channels which partly consist of explicitly sampleable distributions, such as Gaussian sensor noise of digital photographs. To achieve a high encoding rate, we extend the formalisation of stegosystems to capture a technique known as “cover-source switching” in the practical steganography literature. This allows us to construct CCA-secure stegosystems for Gaussian channels using Gaussian preimage sampling techniques borrowed from lattice-based cryptography, which can theoretically achieve an embedding rate of $1/\omega(\log \log \lambda)$ regardless of the min-entropy of the channel.

Our prototype implementation suggests that our scheme is practical, achieving an embedding rate of 24.7% in 24-megapixel RAW images in around 1 minute per image.

1 Introduction

Steganography studies methods to conceal messages within naturally distributed coverttexts, such as texts, images, videos, audio tracks, Internet packets, artificial intelligence generated content and so on. In a nutshell, a stegosystem consists of a (stego)encoder and a (stego)decoder. The encoder aims to hide a message in a stegotext, which is indistinguishable from a sample of natural coverttext from the view of the adversary³, such that a secret-key-holding decoder can decode the stegotext to recover the message. The concept of natural coverttext is modelled via a coverttext channel, which roughly refers to a family of distributions describing what the “next coverttext” will be. In a sense, steganography is a strengthening of encryption: not only the message but the very act of sending it is also hidden.

1.1 Rejection sampling

An established technique which is used in virtually all provably secure stegosystems [HLvA02, vAH04, BC05, BL18] is rejection sampling. Suppose msg is a message to be embedded, $\mathcal{C}$ is a coverttext channel with min-entropy h , and H is a strongly 2-universal hash function. The first step of the stego-encoder is typically to turn msg into a ciphertext ctxt using an encryption scheme with pseudorandom ciphertexts. Then, the ciphertext ctxt is split into γ chunks of ℓ -bit strings $\text{ctxt}_1, \text{ctxt}_2, \dots, \text{ctxt}_\gamma$. A rejection sampling procedure is

³ Also called the warden or the steganalyst.

then performed to embed each ctxt_i : Repeatedly sampling a coverttext stxt_i from $\mathcal{C}$ until $H(\text{stxt}_i) = \text{ctxt}_i$ or until a maximum number N samples have been drawn, i.e. the embedding procedure fails. The coverttexts $\text{stxt} = (\text{stxt}_1, \text{stxt}_2, \dots, \text{stxt}_\gamma)$ returned by the above procedure is then output as the stegotext.

If the underlying encryption is replayable chosen-ciphertext attack (RCCA) secure⁴ [CKN03], Backes and Cachin [BC05] showed that the above construction is replayable chosen-coverttext attack (RCCA) secure for any channel $\mathcal{C}$ with high enough min-entropy, i.e. the construction yields a universal stegosystem.

Looking at the parameters more closely, to achieve security, the min-entropy h needs to be sufficiently larger than ℓ , i.e. $h \geq \ell + \omega(\log \lambda)$ ([BC05, Prop. 7]). For the rejection sampling procedure to be reliable and run in polynomial time, we must have $\ell \leq O(\log \lambda)$ ([BC05, Thm. 6]). This means that the construction only works when the channel has min-entropy at least $\omega(\log \lambda)$, at least $\omega(\log \lambda)$ bits are “wasted” per coverttext and the waste only grows as h increases since ℓ is capped at $O(\log \lambda)$. In other words, the encoding rate $O(\log \lambda)/h$ is at best $o(1)$ and gets worse as the channel becomes more entropic.

1.2 CCA security

The construction of Backes and Cachin does not achieve (full) CCA security, as demonstrated by Berndt and Liśkiewicz [BL18], since an adversary could query the decoding oracle on stegotexts stxt which hash to the same ciphertext ctxt or on permutations of the challenge stegotext stxt^* . In fact, Berndt and Liśkiewicz [BL18] showed a general impossibility of CCA-secure universal stegosystem for channels with memory. More precisely, they showed that even for a class of channels with minimal memory called 0-memoryless channels, where a channel distribution only depends on the length but not the value of the history, there does not exist a stegosystem in the so-called non-look-ahead model that is CCA-secure for all channels in this class. Here, non-look-ahead means that the stegoencoder must commit to the current stegotext symbol before sampling the “next” one.

Conversely, for memoryless channels, i.e. the coverttext distribution is independent of the history of past coverttexts, [BL18] achieved (full) CCA security by further relying on a pseudorandom permutation (PRP) and a collision-resistant hash function H' . Briefly, the idea is to hash the coverttexts $\text{stxt} = (\text{stxt}_1, \text{stxt}_2, \dots, \text{stxt}_\gamma)$ using H' , shuffle them in a way that is determined by the PRP, and embed the PRP secret key and the digest of H' into the stego-text. In this way, perturbations of the challenge stegotext stxt^* are unlikely to agree with the hidden PRP secret key and the hash digest, and hence can be detected and rejected by the stego-decoder. For this technique to apply, it is crucial that the coverttext channel is memoryless or else shuffling the coverttexts would disturb the dependency.

1.3 Dead end for provable steganography?

The above paints a grim picture for provable steganography: Theoretically, there seems to be no good answers to the question of CCA-secure steganography due to the general impossibility for channels with the most minimal memory and the crucial reliance on the memoryless property for the only known construction. Practically, even if we settle for RCCA-security, rejection-sampling-based stegosystems inherently have a low encoding rate especially for channels with high min-entropy.

Even for the best case scenario of low min-entropy memoryless channels, where CCA-security is achievable and with more favourable encoding rates, finding natural channels in this class is not always easy. For example, consider the setting of digital images: If we model the coverttext channel as the distribution of the “next pixel”, then the min-entropy is low (e.g. at most 11 bits for an 11-bit RAW image) but the channel is unlikely to be memoryless – permuting the pixels of a cat image would likely result in an image that does not resemble a cat. In contrast, if we model the channel as the distribution of entire images, then we can heuristically assume that the channel is memoryless (permuting the ordering of multiple cat images may still result in a sequence that is considered natural), but the min-entropy of the channel would be high (e.g. millions of bits for a 24-megapixel image).

⁴ Same as CCA-security, except that the adversary is not allowed to query the decoding oracle on any ciphertext which decrypts to any of the challenge plaintext.

1.4 Our contributions

We make both theoretical and practical contributions to provable steganography.

Circumventing impossibility. Recall that Berndt and Liśkiewicz [BL18] showed the non-existence of CCA-secure steganography for 0-memoryless channels in the non-look-ahead model. We circumvent this impossibility result by dropping the non-look-ahead restriction and constructing a CCA-secure stegosystem for a class of 0-memoryless channels which we call “partially sampleable channels” in the random oracle model. The basic idea is to exploit the existence of an efficient algorithm which samples part of the channel to sample stegotexts in a pseudorandom way inspired by the Fujisaki-Okamoto transformation [FO13].

We note that the non-look-ahead model was regarded as natural in the provable steganography literature [HLvA02, vAH04, BC05, BL18] and all prior constructions work in this model. However, to our knowledge, it was not known how a stegoencoder can exploit the ability to look ahead in any meaningful way, including to circumvent the impossibility result on CCA-security.

Formalisation of cover-source switching. Our generic construction mentioned above still relies on rejection sampling. To obtain a high-rate construction for realistic channels, we turn to the notion of cover-source switching [Bas16b, Bas16a] originating from the field of practical (but not provable) steganography. In more formal language, the idea of cover-source switching is to construct a stegoencoder which has access to a channel $\mathcal{C}$ and which outputs stegotexts which are indistinguishable from samples of another (correlated) channel $\tilde{\mathcal{C}}$. Predominant examples of such a pair of channels considered in the literature [Bas16b, Bas16a, Bas17, DBF18, TBSF19, TBFS19, TBSF21] are digital photographs taken at different ISO settings which have different noise profiles. We extend the formal syntax and definition of CCA-security to capture stegosystems constructed following the cover-source switching paradigm.

High-rate construction for Gaussian channels. Motivated by a generally accepted assumption that ISO noise of digital photographs can be modelled as Gaussian distributions [Bas16a], we construct a CCA-secure stegosystem for correlated Gaussian channels in the random oracle model. Deviating from all existing constructions of provably secure stegosystems, which are built from generic primitives such as universal hash functions and public-key encryption schemes, our construction crucially relies on techniques borrowed from lattice-based cryptography, particularly Gaussian preimage sampling [GPV08, Pei10, MP12]. In a nutshell, our construction involves embedding ciphertexts in a non-black-box manner as Gaussian noise over lattice cosets. By relying on Gaussian preimage sampling, we are no longer constrained by the $O(\log \lambda)/h$ encoding rate of rejection sampling. Instead, we can theoretically achieve an encoding rate of $1/\omega(\log \log \lambda)$ (assuming rate-1 encryption) regardless of how large the min-entropy h of the channel is.⁵

Proof-of-concept implementation. To demonstrate the feasibility of our construction for Gaussian channels, we provide a proof-of-concept implementation in Python. Our concrete choice of the cover channel is RAW images⁶ with $6000 \times 4000 = 24$ megapixels (and thus have high min-entropy) taken with a Sony A6400 camera at ISO100, which are modelled as vectors following a (clipped⁷) discrete Gaussian distribution according to the noise profile estimated in [WYYL24]. Our prototype stegoencoder embeds messages in images which pretend to be ISO200, 400 and 800 images in around 1 minute. The uncompressed size of each image is 48.7 MB regardless of the ISO level. The embedded message sizes (and embedding rates w.r.t. the uncompressed image size) are 3.8 MB (7.9%), 9.4 MB (19.3%) and 12.0 MB (24.7%) respectively. For comparison, existing stegoencoders based on cover-source switching are only practical on small images (e.g. 512×512 pixels) [FPB, CGB19].

⁵ The discussion here is simplified in that a realistic channel may have varying min-entropy over time (or space, e.g. for images).

⁶ For simplicity of the construction, we only formally consider a setting without demosaicing or any out-of-camera post-processing, and discuss how to lift these restrictions informally.

⁷ Values beyond the boundaries are clipped to the boundaries.

2 Technical overview

This work is somewhat unusual in that it puts together techniques from three disjoint communities: Cover-source switching [Bas16b, Bas16a] from practical steganography, formal modelling of provable steganography [HLvA02, vAH04, BC05, BL18] and Gaussian preimage sampling [GPV08, MP12] from lattice-based cryptography. This high-level technical overview aims to assemble these ideas in a coherent and logical way.

2.1 Cover-source switching and Natural Steganography

Cover-source switching is a general paradigm for steganography first proposed by Bas [Bas16b, Bas16a]. Different from the traditional approach of embedding messages through distorting samples obtained from a coartext channel $\mathcal{C}$ (also called a cover-source), Bas proposed to use the difference between two cover-sources $\mathcal{C}$ and $\tilde{\mathcal{C}}$ to embed messages.

In more precise terms, we can interpret Bas’ framework as follows. Consider a stego-encoder $\text{Enc}^{\mathcal{C}}$ which is given oracle-access to a channel $\mathcal{C}$.⁸ On input a message msg , its goal is to produce a stegotext stxt such that it is computationally indistinguishable from a sample drawn from the other channel $\tilde{\mathcal{C}}$ but decodable to msg by a stego-decoder Dec possessing the secret key. We note that this security guarantee is a generalisation of that of traditional stegosystems [HLvA02, vAH04, BC05, BL18] which require the above to hold for $\mathcal{C} = \tilde{\mathcal{C}}$.

While the concept of cover-source switching can in principle be applied to many types of cover-sources, Bas [Bas16b, Bas16a] specifically considered instantiations of cover-source switching steganography for digital images, by exploiting the difference in the sensor noise of cameras at different ISO settings, and called such instantiations “Natural Steganography”. Bas’ [Bas16b, Bas16a] and a sequence of follow-up works [Bas17, DBF18, TBSF19, TBFS19, TBSF21] demonstrated that Natural Steganography could offer better empirical security than competing stegosystems.

2.2 Natural Steganography meets lattice-based cryptography

The (so far empirical) security of Natural Steganography, or cover-source switching steganography in general, crucially depends on our (the stegosystem designer’s) understanding of the cover-source distributions. In the case of Natural Steganography, it has been proposed [Bas16a] and partially verified [WYYL24] that the ISO noise of digital images follows a heteroscedastic (clipped) Gaussian distribution, where the variance depends on the mean of the pixel value (i.e. the “true” intensity of light received by the sensor unit) and the ISO level. In existing Natural Steganography schemes [Bas16b, Bas16a, Bas17, DBF18, TBSF19, TBFS19, TBSF21], the ISO noise model is used to generate stegotexts in the following way: First, the distortion cost of each pixel is calculated based on the noise model. Then, Syndrome Trellis Codes (STCs) are used to (deterministically) embed messages into codewords, which are output as stegotexts [FJF11]. While the above methodology is based on the general principle of cover-source switching, not much can be formally stated about the distribution of the resulting stegotexts, in particular whether it is close to the target coartext distribution $\tilde{\mathcal{C}}$.

In this work, we explore a completely different method inspired by lattice-based cryptography: To directly sample Gaussian noise from lattice cosets encoding (the ciphertexts of) messages. In more detail, consider a simplified setting where images in $\mathcal{C}$ and $\tilde{\mathcal{C}}$ are modelled as vectors following discrete Gaussian distributions $\mathcal{D}_{\mathbb{Z}^p, \mathbf{S}, \mathbf{p}^\dagger}$ and $\mathcal{D}_{\mathbb{Z}^p, \tilde{\mathbf{S}}, \mathbf{p}^\dagger}$ respectively⁹, where $\mathbf{p}^\dagger$ denotes the “true” intensity of the pixels. The only difference is hence the Gaussian parameter $\mathbf{S}$ and $\tilde{\mathbf{S}}$. For concreteness, think of $\mathcal{C}$ as the distribution of ISO100 images capturing a particular scene, and $\tilde{\mathcal{C}}$ as the distribution of ISO800 images¹⁰ capturing the same scene. Let $\mathbf{G} \in \mathbb{Z}_q^{n \times p}$ denote a gadget matrix [MP12] which admits a publicly computable lattice trapdoor. The basic idea is, given a coartext $\mathbf{p} \leftarrow \mathcal{D}_{\mathbb{Z}^p, \mathbf{S}, \mathbf{p}^\dagger}$ sampled from $\mathcal{C}$, to encode a ciphertext ctxt encrypting msg as a

⁸ To capture constructions based on both rejection sampling and Gaussian preimage sampling, we formally allow Enc to have oracle access to both $\mathcal{C}$ and $\tilde{\mathcal{C}}$.

⁹ $\mathcal{D}_{\mathbb{Z}^p, \mathbf{S}, \mathbf{c}}(\mathbf{x}) \propto \exp(-\pi(\mathbf{x} - \mathbf{c})^\top \Sigma^{-1}(\mathbf{x} - \mathbf{c}))$ is the discrete Gaussian distribution centred at $\mathbf{c}$ with support restricted to the integer lattice $\mathbb{Z}^p$ and with $\Sigma = \mathbf{S}\mathbf{S}^\top$. The centre $\mathbf{c}$ is taken as $\mathbf{0}$ if omitted.

¹⁰ The gap between the ISO levels must be large enough for there to be enough space for embedding.

vector $\mathbf{c} \in \mathbb{Z}_q^n$ and sample a new cocontext $\tilde{\mathbf{p}} \leftarrow \mathcal{D}_{\mathbb{Z}^p, \tilde{\mathbf{S}} - \mathbf{S}, \mathbf{p}}$, i.e. a discrete Gaussian with parameter $\tilde{\mathbf{S}} - \mathbf{S}$ and centre $\mathbf{p}$, subject to the constraint $\mathbf{G}\tilde{\mathbf{p}} = \mathbf{c} \bmod q$ using the public trapdoor of $\mathbf{G}$. In an appropriate parameter regime, we can argue that $\tilde{\mathbf{p}}$ sampled as such is statistically indistinguishable from a sample drawn from $\tilde{\mathcal{C}} = \mathcal{D}_{\mathbb{Z}^p, \tilde{\mathbf{S}}, \mathbf{p}^\dagger}$, while the stego-decoder can easily recover ctxt from $\mathbf{c} = \mathbf{G}\tilde{\mathbf{p}} \bmod q$ and then recover the message msg . If the difference $\tilde{\mathbf{S}} - \mathbf{S}$ in Gaussian parameters is sufficiently large, then it can be shown that this basic idea leads to an RCCA-secure stegosystem with a similar security proof as in [BC05].

Let s be the minimum singular value of $\tilde{\mathbf{S}} - \mathbf{S}$. The min-entropy of $\mathcal{D}_{\mathbb{Z}^p, \tilde{\mathbf{S}} - \mathbf{S}, \mathbf{p}}$ is roughly given by $p \log s$. In the ideal case where s is sufficiently large (technically, larger than the smoothing parameter of the kernel lattice of $\mathbf{G}$), then using the above Gaussian preimage sampling techniques, meaning that there is sufficient noise for the entire image, we can embed close to $p \log s / \omega(\log \log \lambda)$ bits (of the ciphertext encrypting the message) into the image. In particular, as the entropy of the channel $\tilde{\mathcal{C}}$ increases, i.e. the Gaussian parameter s increases, the maximum message length increases with it.

2.3 Achieving CCA-security

The above construction is vulnerable to chosen-cocontext attacks. Indeed, given a challenge stegotext $\tilde{\mathbf{p}}^*$, using the public trapdoor of $\mathbf{G}$, the adversary could efficiently sample another cocontext $\tilde{\mathbf{p}}'$ satisfying $\mathbf{G}\tilde{\mathbf{p}}^* \bmod q = \mathbf{G}\tilde{\mathbf{p}}' \bmod q$ and query the decoding oracle on $\tilde{\mathbf{p}}'$ to recover the challenge message. To achieve CCA-security, the stego-encoder must therefore somehow authenticate the stegotext $\tilde{\mathbf{p}}$ such that it is infeasible for the adversary to produce correlated valid stegotexts.

Towards this end, recall that the Fujisaki-Okamoto (FO) transformation [FO13] can be used to turn a CPA-secure public-key encryption (PKE) scheme into a CCA-secure one in the random oracle model. In more detail, the FO transform yields a PKE scheme in the hybrid encryption paradigm: an underlying PKE is used to encrypt a symmetric key, where both the encryption randomness and the symmetric key are derived from a random oracle, and then a symmetric-key encryption (SKE) is used to encrypt the message. Our idea is to treat the stegosystem construction as a hybrid encryption scheme with special PKE and SKE (that output ciphertexts distributed as cocontexts) and apply the FO transform.

In more detail, on input the public key pk , an ISO100 image (for concreteness, or generally a small-parameter Gaussian) and a message msg , our stego-encoder works as follows: We divide the pixels of the ISO100 image into chunks, each consisting of two parts, a large “head” and a small “tail”. We use heads to denote the collection of all heads and tails the collection of all tails. The encoder first SKE-encrypts $(\text{msg}, \text{tails})$ using the key $G_1(\text{pk}, \text{ek})$, where G_1 is a random oracle and ek is a random ephemeral key, and uses Gaussian preimage sampling to embed such a ciphertext into heads to give $\widetilde{\text{heads}}$. Then, the encoder derives some randomness $G_2(\text{pk}, \text{ek}, \widetilde{\text{heads}})$ using another random oracle G_2 , and uses it to PKE-encrypt ek and embed the ciphertext into tails to give $\widetilde{\text{tails}}$.

Correspondingly, on input $(\widetilde{\text{heads}}, \widetilde{\text{tails}})$, the stego-decoder works by first performing a trial decryption to obtain $(\text{msg}, \text{tails})$, and then re-encodes it to obtain $\widetilde{\text{tails}}'$. If $\widetilde{\text{tails}} = \widetilde{\text{tails}}'$, then it accepts the stegotext and outputs msg , otherwise it rejects the stegotext.

Observe that the above construction only makes use of the property that the channel $\tilde{\mathcal{C}}$ is “partially sampleable”, in that there exists an explicit efficient algorithm for sampling the tails, which is intended to be a small part of the channel output. Indeed, in Section 5.2 we present a generic construction of CCA-secure public-key stegosystems from a standalone primitive that we call “Natural Steganographic Encoding” (NSE). We can then instantiate the NSE with rejection sampling for arbitrary partially sampleable channels (also when $\mathcal{C} = \tilde{\mathcal{C}}$) or with Gaussian preimage sampling for Gaussian channels.

2.4 Handling realistic image distributions

Turning to the practical side, while the construction sketched above achieves correctness and CCA-security for discrete Gaussian channels $\mathcal{C} = \mathcal{D}_{\mathbb{Z}^p, \mathbf{S}, \mathbf{p}^\dagger}$ and $\tilde{\mathcal{C}} = \mathcal{D}_{\mathbb{Z}^p, \tilde{\mathbf{S}}, \mathbf{p}^\dagger}$, where $\tilde{\mathbf{S}} - \mathbf{S}$ is large enough, it is unlikely to allow correct recovery of the message for a more realistic modelling of image distributions. There are at least two subtleties to be taken into account:

1. It may happen for some dimensions the noise level represented by $\tilde{\mathbf{S}} - \mathbf{S}$ is not large enough for statistically indistinguishable preimage sampling. For example, bright areas in an image tend to have lower noise levels than dark areas. To not harm security, these low-noise pixels should be ignored when embedding messages, i.e. they should be sampled directly according to $\tilde{\mathcal{C}}$. However, there is no way for the stego-encoder to inform the stego-decoder about which pixel is being ignored.
2. A step that is omitted from the above is that the final output of the stego-encoder is clipped, i.e. if a pixel value exceeds some predefined boundaries, it is set to the value of the closest boundary.

Since the stego-decoder sketched above is not robust to errors, failure to embed in a single pixel may destroy the entire stegotext.

To introduce robustness, our idea is to first encode $(\text{msg}, \text{tails})$, together with an “error report” aux of which parts in tails are directly sampled according to $\tilde{\mathcal{C}}$, using an error-correcting code (ECC). During trial decryption, the stego-decoder may recover a perturbed ECC codeword with perturbation due to the clipping error mentioned above. As long as the number of perturbed codeword symbols is within the error correcting capability of the ECC, the stego-decoder could still recover $(\text{msg}, \text{tails}, \text{aux})$ in full and perform the re-encoding step (which reproduces the same clipping errors if any).

On top of the above sources of errors, we note that other image processing operations such as compression, quantisation, colour correction, etc. may also affect the reliability or even security of the stegosystem. Dealing with these issues is possible but out of the scope of this work, and we focus on RAW images which do not involve these complications.

3 Preliminaries

Denote $\mathbb{N} := \{0, 1, 2, \dots\}$ and let $\lambda \in \mathbb{N}$ be the security parameter. The set of positive polynomials and negligible functions in λ are denoted $\text{poly}(\lambda)$ and $\text{negl}(\lambda)$ respectively. For positive $n \in \mathbb{N}$, write $[n] := \{1, \dots, n\}$. For a finite set S , write $x \leftarrow \$ S$ for sampling a uniformly random element. If X is a (discrete) distribution over some (countable) set S , we write $X \sim S$. We also write $x \leftarrow \$ X$ for sampling according to the distribution X . For two distributions X and Y , we write $X \approx_s Y$ to mean that their statistical distance is upper-bounded by $\text{negl}(\lambda)$.

In the formal definition of steganography we will handle “histories” which are strings $\text{hist} = c_0 \| c_1 \| \dots \| c_{\ell-1}$ where each c_i belongs to some (possibly different) alphabet. We write $|\text{hist}| = \ell$ for the length of the string in number of symbols. For two strings hist_0 and hist_1 we write $\text{hist}_0 \| \text{hist}_1$ for their concatenation.

3.1 Lattices and discrete Gaussians

Lattices. An m -dimensional lattice Λ is a discrete additive subgroup of $\mathbb{R}^m$. A matrix $\mathbf{B} \in \mathbb{R}^{m \times m}$ is a basis of a full-rank m -dimensional lattice Λ , if the columns of $\mathbf{B}$ are linearly-independent and $\mathbf{B}$ generates Λ , i.e. it holds $\Lambda = \mathbf{B} \cdot \mathbb{Z}^m = \{\mathbf{B} \cdot \mathbf{z} : \mathbf{z} \in \mathbb{Z}^m\}$. For any $\mathbf{A} \in \mathbb{Z}_q^{n \times m}$ and $\mathbf{v} \in \mathbb{Z}_q^n$, we use the shorthand

$$\Lambda_q^\perp(\mathbf{A}) := \{\mathbf{u} \in \mathbb{Z}^m : \mathbf{A} \cdot \mathbf{u} = \mathbf{0} \bmod q\}, \quad \Lambda_q^\mathbf{v}(\mathbf{A}) := \{\mathbf{u} \in \mathbb{Z}^m : \mathbf{A} \cdot \mathbf{u} = \mathbf{v} \bmod q\}.$$

Gaussian function. For a positive semi-definite¹¹ matrix $\Sigma \in \mathbb{R}^{m \times m}$, the Gaussian function with width parameter Σ and centre $\mathbf{c} \in \mathbb{R}^m$ is $\rho_{\sqrt{\Sigma}, \mathbf{c}}(\mathbf{x}) := \exp(-\pi(\mathbf{x} - \mathbf{c})^\top \Sigma (\mathbf{x} - \mathbf{c}))$ for all $\mathbf{x} \in \mathbb{R}^m$, where $\sqrt{\Sigma}$ denotes the matrix $\mathbf{S} \in \mathbb{R}^{m \times m}$ such that $\mathbf{S}\mathbf{S}^\top = \Sigma$.¹² The subscripts are taken to be $\sqrt{\Sigma} = \mathbf{I}$ and $\mathbf{c} = \mathbf{0}$ when omitted. If $\Sigma = \frac{1}{\chi^2} \mathbf{I}$ for some $\chi \in \mathbb{R}^+$, we simply write $\rho_{\chi, \mathbf{c}}$.

Discrete Gaussians. The discrete Gaussian distribution $\mathcal{D}_{\Lambda, \Sigma, \mathbf{c}}$ over a lattice Λ is defined as $\mathcal{D}_{\Lambda, \sqrt{\Sigma}, \mathbf{c}}(\mathbf{x}) = \frac{\rho_{\sqrt{\Sigma}, \mathbf{c}}(\mathbf{x})}{\rho_{\sqrt{\Sigma}, \mathbf{c}}(\Lambda)}$, for all $\mathbf{x} \in \Lambda$, i.e. it is proportional to $\rho_{\sqrt{\Sigma}, \mathbf{c}}(\mathbf{x})$ but normalised to make it a probability distribution.

¹¹ A symmetric $\Sigma \in \mathbb{R}^{m \times m}$ is positive semi-definite if $\mathbf{x}^\top \Sigma \mathbf{x} \geq 0$ for all non-zero $\mathbf{x} \in \mathbb{R}^m$.

¹² Such a root $\mathbf{S}$ can be computed efficiently, e.g. by the Cholesky decomposition.

Smoothing parameter [MR04]. For $\epsilon > 0$, the ϵ -smoothing parameter of a lattice Λ is $\eta_\epsilon(\Lambda) := \inf\{s \in \mathbb{R}^+ : \rho_{1/s}(\Lambda^* \setminus \{\mathbf{0}\}) \leq \epsilon\}$, where Λ^* is the dual lattice of Λ .

Lemma 1 ([MP12, Lem. 2.3]). *Let $\Lambda \subset \mathbb{R}^m$ be a lattice with basis $\mathbf{B}$, and let $\epsilon > 0$. We have $\eta_\epsilon(\Lambda) \leq \|\tilde{\mathbf{B}}\| \cdot \sqrt{\ln(2m(1+1/\epsilon))/\pi}$, where $\tilde{\mathbf{B}}$ is the Gram-Schmidt orthogonalisation of $\mathbf{B}$. In particular, for $m \in \text{poly}(\lambda)$ and any function $f(\lambda) \geq \omega(\sqrt{\log \lambda})$, there is a negligible $\epsilon = \epsilon(\lambda)$ such that $\eta_\epsilon(\Lambda) \leq \|\tilde{\mathbf{B}}\| \cdot f(\lambda)$.*

Lemma 2 ([PR06, Lem. 2.10]). *Let $\Lambda \subseteq \mathbb{R}^m$ be an m -dimensional full-rank lattice. For any $\epsilon \in (0, 1)$, $s \geq 2 \cdot \eta_\epsilon(\Lambda)$, $\mathbf{x} \in \Lambda$, and $\mathbf{c} \in \mathbb{R}^m$, it holds $\mathcal{D}_{\Lambda, s, \mathbf{c}}(\mathbf{x}) \leq 2^{-m} \cdot \frac{1+\epsilon}{1-\epsilon}$.*

3.2 Gadget matrix and Gaussian preimage sampling

Gadget matrix [MP12]. For any positive integers n and $\beta \leq q$, the gadget vector is $\mathbf{g}_{\beta, q}^\top := (1, 2, \dots, 2^k)$ where $k = \lceil \log_\beta q \rceil$, and the gadget matrix is $\mathbf{G}_{n, \beta, q} := \mathbf{I}_n \otimes \mathbf{g}_{\beta, q}^\top$. The operator $\mathbf{G}_\beta^{-1}(\cdot)$ is the deterministic β -ary decomposition. The subscript q is by default the modulus and omitted, the subscripts n and/or β are dropped when they are clear from the context. We have $\mathbf{G}\mathbf{G}^{-1}(\mathbf{x}) = \mathbf{x} \bmod q$ for any $\mathbf{x} \in \mathbb{Z}_q^n$.

Lemma 3 ([MP12, Thm. 4.1]). *For any integers $q \geq 2$, $n \geq 1$, $\beta \geq 2$, $m = n \lceil \log_\beta q \rceil$, there is a known basis $\mathbf{S} \in \mathbb{Z}^{m \times m}$ of $\Lambda_q^\perp(\mathbf{G}_{n, \beta})$ with $\|\tilde{\mathbf{S}}\| \leq \sqrt{1 + \beta^2}$, where $\tilde{\mathbf{S}}$ is the Gram-Schmidt orthogonalisation of $\mathbf{S}$.*

Lemma 4 (Preimage sampling [MP12]). *Let $n, \beta, q \in \mathbb{N}$ with $\beta \leq q$. There exists a PPT algorithm SampPre with the syntax $\mathbf{u} \leftarrow \text{SampPre}(n, \beta, \mathbf{v}, s', \mathbf{c})$, such that for any $\mathbf{v} \in \mathbb{Z}_q^n$, any $\mathbf{c} \in \mathbb{R}^n$, and any $s' \geq \sqrt{1 + \beta^2} \cdot \omega(\sqrt{\log \lambda})$, the following hold:*

1. $\mathbf{G}_{n, \beta} \cdot \mathbf{u} = \mathbf{v} \bmod q$ with probability 1 for $\mathbf{u} \leftarrow \text{SampPre}(n, \beta, \mathbf{v}, s', \mathbf{c})$.
2. $\text{SampPre}(n, \beta, \mathbf{v}, s', \mathbf{c})$ and $\mathcal{D}_{\Lambda_q^\perp(\mathbf{G}_{n, \beta}), s', \mathbf{c}}$ are statistically close in λ .

3.3 Other primitives and tools

We give definitions of standard cryptographic primitives, e.g. public-key encryption (PKE), pseudorandom functions (PRF) and universal hash functions in Appendix A. We define error-correcting codes (ECCs) with a syntax that is convenient for our purposes.

Definition 1 (Error-correcting code). *An error-correcting code (ECC) with message space $\mathcal{M}$ and codeword space Γ_{ECC}^N is a tuple of PPT algorithms $\text{ECC} = (\text{Enc}, \text{Dec})$ with the following syntax:*

$\text{cw} \leftarrow \text{Enc}(\text{msg})$: *The encoding algorithm takes as input a message $\text{msg} \in \mathcal{M}$ and returns a codeword $\text{cw} \in \Gamma_{\text{ECC}}^N$.*
 $\text{msg}' \leftarrow \text{Dec}(\text{cw})$: *The decoding algorithm takes as input a codeword $\text{cw} \in \Gamma_{\text{ECC}}^N$ and returns a message $\text{msg}' \in \mathcal{M}$.*

An ECC is said to be δ_{ECC} -error-correcting, if for any $\text{msg} \in \mathcal{M}$ and any word $\text{cw}^ \in \Gamma_{\text{ECC}}^N$ it holds that*

$$\text{dist}(\text{Enc}(\text{msg}), \text{cw}^*) \leq \delta_{\text{ECC}} \implies \text{Dec}(\text{cw}^*) = \text{msg}$$

where dist is the Hamming distance over Γ_{ECC} . We say that ECC is correct if it is 0-error-correcting, i.e. without any error-correcting capability.

4 Formalising steganography with cover-source switching

To formally discuss steganography, we need the notion of *channels*. Below, we state a slightly generalised version which allows different alphabets.

Definition 2 (Channel). A channel $\mathcal{C} = (\mathcal{C}_{\text{hist}})_{\text{hist}}$ is a family of distributions $\mathcal{C}_{\text{hist}}$ over some alphabets Σ_{hist} indexed by finite strings $\text{hist} \in \Sigma_{\epsilon} \times \Sigma_{\text{hist}_1} \times \dots \times \Sigma_{\text{hist}_{|\text{hist}|-1}} = \prod_{i=0}^{|\text{hist}|-1} \Sigma_{\text{hist}_i}$ where hist_i is the i -prefix of hist . A channel $\mathcal{C}$ is said to be memoryless if $\mathcal{C}_{\text{hist}} = \mathcal{C}_{\text{hist}'}$ for all $\text{hist}, \text{hist}'$. It is said to be 0-memoryless if $\mathcal{C}_{\text{hist}} = \mathcal{C}_{\text{hist}'}$ whenever $|\text{hist}| = |\text{hist}'|$.

We often omit the domain $\prod_{i=0}^{|\text{hist}|-1} \Sigma_{\text{hist}_i}$ for hist when quantifying over it for conciseness. When a channel $\mathcal{C} = (\mathcal{C}_{\text{hist}})_{\text{hist}}$ is 0-memoryless, we can more simply write $\mathcal{C} = (\mathcal{C}_i)_{i \in \mathbb{N}}$ where each $\mathcal{C}_i$ is a distribution over some alphabet Σ_i without ambiguity with the understanding that $\mathcal{C}_i = \mathcal{C}_{\text{hist}}$ and $\Sigma_i = \Sigma_{\text{hist}}$ for $|\text{hist}| = i$.

Definition 3 (Algorithms with oracle access to channel). For an oracle-aided algorithm $\mathcal{A}$, we write $\mathcal{A}^{\mathcal{C}}$ if $\mathcal{A}$ has oracle access to $\mathcal{C}_{\text{hist}}$ for all hist . We write $\mathcal{A}^{\mathcal{C}_{\geq \text{hist}^*}}$ if $\mathcal{A}$ has oracle access to $\mathcal{C}_{\text{hist}}$ for all hist having hist^* as a prefix.

Intuitively, the definition of channels models the following stochastic process:

Given the current history hist , sample $c \leftarrow \mathcal{C}_{\text{hist}}$ then extend the history by c , i.e. $\text{hist} := \text{hist}||c$. Repeat.

Roughly, a stegosystem is said to be secure if the stegotexts it produces are indistinguishable from coverttexts sampled from $\mathcal{C}$. That is, it models the following alternative process:

Given the current history hist , sample $c \leftarrow \mathcal{C}_{\text{hist}}$, post-process the sample with some steganographic algorithm f_x to obtain $c' = f_x(c)$ which embeds some hidden information x , then extend the history by c' . Repeat.

The provable steganography literature typically considers the so-called non-look-ahead model, which roughly means that the steganographic algorithm never “looks ahead” to future distributions. As far as we are aware, the concept of non-look-ahead has only been described informally and no precise definition is explicitly given in the literature. We attempt to give a formal definition below along with a few other reasonable models for channel query patterns.

Definition 4 (Channel query patterns). For an oracle-aided algorithm $\mathcal{A}$ modelled as a Turing or random access machine, a history hist and an auxiliary aux , let $\text{str}_1, \text{str}_2, \dots$ (interpreted as nodes of a $|\Sigma|$ -ary tree) index the sequence of distributions that $\mathcal{A}^{\mathcal{C}}(\text{hist}, \text{aux})$ queries. We say that $\mathcal{A}^{\mathcal{C}}$ has the following query patterns if the corresponding properties are satisfied for all $(\text{hist}, \text{aux})$:

- Random access (default): $\text{str}_1, \text{str}_2, \dots$ are in no particular order.
- Tree traversal: $\text{str}_1 = \text{hist}$ and each str_i either belongs to or is a child of a node in $\{\text{str}_1, \dots, \text{str}_{i-1}\}$.
- Pathfinding: $\text{str}_1 = \text{hist}$ and each str_i is either equal to or is a child of str_{i-1} .
- Non-look-ahead: Same as pathfinding, but additionally the output tape out of $\mathcal{A}^{\mathcal{C}}$ is append-only and $\text{out} = \text{str}_{i-1}$ when $\mathcal{C}_{\text{str}_i}$ is queried.

The above query patterns are presented in an order which is increasingly restrictive. However, these different restrictions collapse to fewer ones (or, in some sense, some restrictions are lifted) when restricting to special classes of channels: If $\mathcal{C}$ is a memoryless channel, then all query patterns are equivalent. If $\mathcal{C}$ is 0-memoryless, then all query patterns except non-look-ahead are equivalent.

4.1 Formalising cover-source switching

In existing formal definitions of steganography [HLvA02, vAH04, BC05, BL18], a stegosystem is defined relative to a *single* channel $\mathcal{C}$ which models the distribution of coverttexts. Variations of the definition considered in the literature, such as symmetric-key vs. public-key, chosen-plaintext attack (CPA) vs. chosen-coverttext attack (CCA), general vs. memoryless channels, etc. do not alter this fact, i.e. stegosystems being defined relative to a single $\mathcal{C}$. As such, these definitions do not seem expressive enough to capture stegosystems constructed using the technique of cover-source switching [Bas16b, Bas16a].

To recall, the general principle of cover-source switching [Bas16b, Bas16a] is, given samples from a channel $\mathcal{C}$, to produce stegotexts which are indistinguishable from coverttexts sampled from another (possibly correlated) channel $\tilde{\mathcal{C}}$. We formalise cover-source switching as follows. Consider two channels $\mathcal{C} = (\mathcal{C}_{\text{hist}})_{\text{hist}}$ and $\tilde{\mathcal{C}} = (\tilde{\mathcal{C}}_{\text{hist}})_{\text{hist}}$ over the same alphabets $(\Sigma_{\text{hist}})_{\text{hist}}$ which are correlated in the sense that the distributions $\mathcal{C}_{\text{hist}}$ and $\tilde{\mathcal{C}}_{\text{hist}}$ can be correlated for some or all hist . This setting intuitively captures the following stochastic process:

Given the current history hist , either sample $c \leftarrow \mathcal{C}_{\text{hist}}$ then extend the history by c or sample $c' \leftarrow \tilde{\mathcal{C}}_{\text{hist}}$ then extend the history by c' . Repeat.

Cover-source switching means that instead of designing steganographic algorithm f_x so that $f_x(\mathcal{C}_{\text{hist}}) \approx \mathcal{C}_{\text{hist}}$, we make it so that $f_x(\mathcal{C}_{\text{hist}}) \approx \tilde{\mathcal{C}}_{\text{hist}}$. The single-channel case can be seen as a specialisation by setting $\mathcal{C} = \tilde{\mathcal{C}}$.

Below, we extend the definition of stegosystems to be relative to two channels $\mathcal{C}$ and $\tilde{\mathcal{C}}$ to capture cover-source switching.¹³ We work over the public-key setting; the definition can be easily adapted to the symmetric-key setting.

Definition 5 (Public-key stegosystem). *A public-key stegosystem (PKS) for a message space $\mathcal{M}$ consists of PPT (oracle-aided) algorithms (KGen, Enc, Dec) with the following syntax:*

(pk, sk) $\leftarrow$ KGen(1^λ): The key generation algorithm outputs a pair of public and secret keys (pk, sk), where pk implicitly defines a coverttext/stegotext space $\mathcal{S}$.
 stxt $\leftarrow$ Enc $^{\mathcal{C}_{\geq \text{hist}}, \tilde{\mathcal{C}}_{\geq \text{hist}}}(\text{pk}, \text{msg}, \text{hist})$: The encoding algorithm, on input the public key pk, a message $\text{msg} \in \mathcal{M}$ and some history hist , provided with oracle-access to some channels $\mathcal{C}_{\geq \text{hist}}$ and $\tilde{\mathcal{C}}_{\geq \text{hist}}$, outputs a stegotext $\text{stxt} \in \mathcal{S}$.
 msg' $\leftarrow$ Dec(sk, stxt): On input a secret key sk and a stegotext stxt, the decoding algorithm outputs a message $\text{msg}' \in \mathcal{M}$.

We say that the PKS is non-look-ahead (resp. pathfinding) if Enc $^{\mathcal{C}, \tilde{\mathcal{C}}}$ is non-look-ahead (resp. pathfinding).

Definition 6 (Correctness). *Let $\mathcal{C}, \tilde{\mathcal{C}}$ be channels over some common alphabet. A PKS for a message space $\mathcal{M}$ is $(\mathcal{C}, \tilde{\mathcal{C}})$ -correct, if for all $\lambda \in \mathbb{N}$, (pk, sk) $\in$ KGen(1^λ), $\text{msg} \in \mathcal{M}$ and possible hist , it holds that*

$$\Pr \left[\text{Dec}(\text{sk}, \text{Enc}^{\mathcal{C}_{\geq \text{hist}}, \tilde{\mathcal{C}}_{\geq \text{hist}}}(\text{pk}, \text{msg}, \text{hist})) = \text{msg} \right] \geq 1 - \text{negl}(\lambda).$$

The above definitions of syntax and correctness are conceptually identical to those in the literature except for the introduction of $\tilde{\mathcal{C}}$ and syntactical changes. Below, we give the corresponding definition of chosen-plaintext attack (CPA) and chosen-coverttext attack (CCA) security also parametrised by two channels $(\mathcal{C}, \tilde{\mathcal{C}})$ instead of one, where an adversary has oracle-access to both $\mathcal{C}$ and $\tilde{\mathcal{C}}$. The stego-encoder Enc with oracle-access to $\mathcal{C}$ and $\tilde{\mathcal{C}}$ is required to produce stegotexts stxt^* indistinguishable from samples from $\tilde{\mathcal{C}}$. By setting $\mathcal{C} = \tilde{\mathcal{C}}$, we recover CPA/CCA security considered in prior works.

Definition 7 (CPA/CCA-security). *Let $\mathcal{C}, \tilde{\mathcal{C}}$ be channels and $0 \leq \epsilon(\lambda) \leq 1$. A PKS Π is said to be $(\mathcal{C}, \tilde{\mathcal{C}}, \epsilon)$ -IND-CCA secure, if for all $\lambda \in \mathbb{N}$,*

$$\left| \Pr \left[\text{Exp}_{\Pi, \mathcal{A}, \mathcal{C}, \tilde{\mathcal{C}}}^0(1^\lambda) = 1 \right] - \Pr \left[\text{Exp}_{\Pi, \mathcal{A}, \mathcal{C}, \tilde{\mathcal{C}}}^1(1^\lambda) = 1 \right] \right| \leq \epsilon$$

¹³ We also introduce syntactical and terminology changes to better align the definitions with e.g. those of public-key encryption.

$\text{Exp}_{\Pi, \mathcal{A}, \mathcal{C}, \tilde{\mathcal{C}}}^b(1^\lambda)$	$\text{DecO}(\text{stxt})$
$(\text{pk}, \text{sk}) \leftarrow \text{KGen}(1^\lambda)$	if $\text{stxt} = \text{stxt}^*$ then
$(\text{hist}^*, \text{msg}^*) \leftarrow \mathcal{A}^{\mathcal{C}, \tilde{\mathcal{C}}, \text{DecO}}(\text{pk})$	return $\perp$
$\text{stxt}^* \leftarrow \text{Enc}_{\geq \text{hist}^*, \tilde{\mathcal{C}}_{\geq \text{hist}^*}}^{\mathcal{C}}(\text{pk}, \text{msg}^*, \text{hist}^*)$	$\text{msg} \leftarrow \text{Dec}(\text{sk}, \text{stxt})$
$\ell^* := \text{stxt}^* $	return msg
if $b = 1$ then	
for $i \in [\ell^*]$ do $s_{i+1} \leftarrow \$ \tilde{\mathcal{C}}_{\text{hist}^* \ s_1 \ \dots \ s_i}$	
$\text{stxt}^* := (s_1 \ \dots \ s_{\ell^*})$ <i>// overwrite stxt*</i>	
$b' \leftarrow \mathcal{A}^{\mathcal{C}, \tilde{\mathcal{C}}, \text{DecO}}(\text{stxt}^*)$	
return b'	

Fig. 1. $(\mathcal{C}, \tilde{\mathcal{C}})$ -CCA security experiment for PKS Π .

holds for any PPT $\mathcal{A}$, where $\text{Exp}_{\Pi, \mathcal{A}, \mathcal{C}, \tilde{\mathcal{C}}}^b$ is defined in Fig. 1. A PKS Π is said to be $(\mathcal{C}, \tilde{\mathcal{C}}, \epsilon)$ -IND-CPA secure if the above holds for any PPT $\mathcal{A}$ without access to the decoding oracle DecO . We omit ϵ if $\epsilon \leq \text{negl}(\lambda)$.

Informally, we may say that a PKS Π is a stegosystem based on cover-source switching if it is $(\mathcal{C}, \tilde{\mathcal{C}})$ -correct and $(\mathcal{C}, \tilde{\mathcal{C}}, \epsilon)$ -secure.

Remark 1 (On non-negligible ϵ). We note that in a practical system a PKS may not be secure in the cryptographic sense, i.e. we may have $\epsilon > \text{negl}(\lambda)$. In the latter case, a PKS should still be cryptographically secure as a PKE scheme, i.e. $(\text{KGen}, \text{Enc}, \text{Dec})$ should satisfy the usual IND-CCA security as a PKE. In other words, an adversary could distinguish the stegotexts of two different messages only with negligible probability, even though it may distinguish stegotexts from coverttexts with small (say < 0.01) but non-negligible probability.

Remark 2 (Channel-aware decoders). In our definitions, we require stego-decoders Dec to be correct even without oracle-access to $(\mathcal{C}, \tilde{\mathcal{C}})$. In general, one could consider stego-decoders Dec which are given oracle-access to $\mathcal{C}$ and/or $\tilde{\mathcal{C}}$.

We measure the efficiency of a stegosystem by its rate, which measures the cost of sending a message in plain relative to sending it as encoded in a stegotext.

Definition 8 (Rate). The rate of a PKS Π for the message space $\mathcal{M}$ and stegotext space $\mathcal{S}$ is $\log |\mathcal{M}| / \log |\mathcal{S}|$.

Berndt and Liškiewicz [BL18] showed that it is impossible for a non-look-ahead PKS to be $\mathcal{C}$ -correct and $\mathcal{C}$ -IND-CCA-security for any 0-memoryless channel $\mathcal{C}$. On the other hand, they showed that it is possible for a non-look-ahead PKS to be $\mathcal{C}$ -correct and $\mathcal{C}$ -IND-CCA-secure for any memoryless channel $\mathcal{C}$. Looking ahead, we will show that if we drop the non-look-ahead requirement, then there exists an (arguably natural) class of 0-memoryless channels, for which it is possible for a PKS to be both $\mathcal{C}$ -correct and $\mathcal{C}$ -IND-CCA-secure for all $\mathcal{C}$ in this class.

5 CCA-secure PKS for 0-memoryless channels

We construct a (pathfinding) PKS which is $(\mathcal{C}, \tilde{\mathcal{C}})$ -correct and $(\mathcal{C}, \tilde{\mathcal{C}})$ -IND-CCA-secure for specialised channels, including what we call partially sampleable channels and twin-Gaussian channels. The former is formalised in Section 5.3 and provides a way to circumvent the impossibility of Berndt and Liškiewicz [BL18] when $\mathcal{C} = \tilde{\mathcal{C}}$; the latter is formalised in Section 5.4 and achieves both improved asymptotic and practical embedding rate when $\mathcal{C} \neq \tilde{\mathcal{C}}$ and $\tilde{\mathcal{C}}$ has high entropy.

To ease notation, we consider the simplification where the channels $\mathcal{C}, \tilde{\mathcal{C}}$ have a common alphabet Σ independent of hist , while noting that all results generalise to the general case with differing alphabets.

5.1 Natural steganographic encodings

The “non-cryptographic” part of our construction is based on the idea of natural steganography [Bas16b, Bas16a, Bas17, DBF18], where the stegoencoder takes as input a sample from $\mathcal{C}$ and attempts to mimic a sample of $\tilde{\mathcal{C}}$ while embedding hidden information. We abstract such a methodology as a standalone primitive that we call *natural steganographic encoding* (NSE).

We consider channels $\mathcal{C}, \tilde{\mathcal{C}}$ with a common alphabet Σ that can be split into a product $\Sigma = \mathbb{H} \times \mathbb{T}$ consisting of two alphabets called the head space and the tail space respectively. At a high level, the heads and tails will function as the symmetric-key and public-key encryption schemes in the Fujisaki-Okamoto (FO) transform [FO13] respectively, where plaintext will be encoded in the heads and the symmetric secret key will be public-key-encrypted and encoded in the tails using pseudorandomness.

Definition 9 (Natural steganographic encoding (NSE)). *Let $\mathcal{C}, \tilde{\mathcal{C}}$ be 0-memoryless channels with common alphabet $\Sigma = \mathbb{H} \times \mathbb{T}$. A natural steganographic encoding (NSE) for channels $(\mathcal{C}, \tilde{\mathcal{C}})$ with length parameter N , head message alphabet $\mathcal{M}_H$ and tail message alphabet $\mathcal{M}_T$ is a tuple of PPT algorithms*

$$\text{NSE} = (\text{EmbH}, \text{EmbT}, \text{ExtH}, \text{ExtT}, \text{ChkT})$$

with the syntax:

- $\widetilde{\text{head}}_i \leftarrow \text{EmbH}^{\tilde{\mathcal{C}}^{\ell+i}}(i, \text{head}_i, \text{msg}_i)$: On input an index $i \in \mathbb{N}$, a head $\text{head}_i \in \mathbb{H}$ and a message $\text{msg}_i \in \mathcal{M}_H$, and given oracle access to the channel $\tilde{\mathcal{C}}$, return an embedded head $\widetilde{\text{head}}_i \in \mathbb{H}$.
- $\widetilde{\text{tail}}_i \leftarrow \text{EmbT}(i, \text{tail}_i, \text{msg}_i, \text{aux})$: On input an index $i \in \mathbb{N}$, a tail $\text{tail}_i \in \mathbb{T}$, a message $\text{msg}_i \in \mathcal{M}_T$ and some auxiliary tail information aux , return an embedded tail $\widetilde{\text{tail}}_i \in \mathbb{T}$.
- $\text{msg}_i \leftarrow \text{ExtH}(\widetilde{\text{head}}_i)$: Deterministically extract $\text{msg}_i \in \mathcal{M}_H$ from $\widetilde{\text{head}}_i \in \mathbb{H}$.
- $\text{msg}_i \leftarrow \text{ExtT}(\widetilde{\text{tail}}_i)$: Deterministically extract $\text{msg}_i \in \mathcal{M}_T$ from $\widetilde{\text{tail}}_i \in \mathbb{T}$.
- $\text{aux} \leftarrow \text{ChkT}(\text{tails})$: Generate auxiliary information aux given a tuple of tails $\text{tails} \in \mathbb{T}^N$.

An NSE with length parameter N is expected to satisfy the following properties:

- (Head Embedding δ -Correctness.) For any $\ell \in \mathbb{N}$, $\text{head}_1, \dots, \text{head}_N \in \mathbb{H}$, $\text{msg}_1, \dots, \text{msg}_N \in \mathcal{M}_H$, it holds that

$$\Pr \left[|\{i \in [N] : \text{ExtH}(\text{EmbH}^{\tilde{\mathcal{C}}^{\ell+i}}(\ell + i, \text{head}_i, \text{msg}_i)) \neq \text{msg}_i\}| > \delta \right] \leq \text{negl}(\lambda).$$

We simply say that NSE has correct head embedding if $\delta = 0$.

- (Tail Embedding τ -Correctness.) For any $\ell, i \in \mathbb{N}$, $\text{tails} = (\text{tail}_i)_{i \in [N]} \in \mathbb{T}^N$, $\text{msg}_i \in \mathcal{M}_T$, it holds that

$$\Pr[\text{ExtT}(\text{EmbT}(\ell + i, \text{tail}_i, \text{msg}_i, \text{ChkT}(\text{tails}))) \neq \text{msg}_i] \leq \tau.$$

We simply say that NSE has correct tail embedding if $\tau \leq \text{negl}(\lambda)$.

- $((\mathcal{C}, \tilde{\mathcal{C}}, \epsilon)$ -Indistinguishability.) For any $\ell \in \mathbb{N}$, the statistical distance between the following distributions is at most ϵ :

$$\left\{ \begin{array}{l} (\widetilde{\text{head}}_i, \widetilde{\text{tail}}_i)_{i \in [N]} \\ (\text{head}_i, \text{tail}_i)_{i \in [N]} \leftarrow \mathcal{C}_\ell \times \dots \times \mathcal{C}_{\ell+N-1} \\ \text{msg} = (\text{msg}_i)_{i \in [N]} \leftarrow \mathcal{M}_H^N \\ \text{msg}_i \leftarrow \text{ExtT}(\widetilde{\text{tail}}_i) \leftarrow \mathcal{M}_T^N \\ \text{aux} \leftarrow \text{ChkT}((\text{tail}_i)_{i \in [N]}) \\ \widetilde{\text{head}}_i \leftarrow \text{EmbH}^{\tilde{\mathcal{C}}^{\ell+i}}(\ell + i, \text{head}_i, \text{msg}_i) \quad \forall i \in [N] \\ \widetilde{\text{tail}}_i \leftarrow \text{EmbT}(\ell + i, \text{tail}_i, \text{msg}_i, \text{aux}) \quad \forall i \in [N] \end{array} \right\} \approx_\epsilon \left\{ \begin{array}{l} (\widetilde{\text{head}}_i, \widetilde{\text{tail}}_i)_{i \in [N]} \\ (\widetilde{\text{head}}_i, \widetilde{\text{tail}}_i)_{i \in [N]} \leftarrow \tilde{\mathcal{C}}_\ell \times \dots \times \tilde{\mathcal{C}}_{\ell+N-1} \end{array} \right\}. \quad (1)$$

- (*Tail γ -Spreadness.*) For any $i \in [N]$, $\text{tails} = (\text{tail}_i)_{i \in [N]} \in \mathbb{T}^N$, $\text{msgt} \in \mathcal{M}_T$, $\text{aux} \in \text{ChkT}(\text{tails})$, it holds that

$$H_\infty(\text{EmbT}(i, \text{tail}_i, \text{msgt}, \text{aux})) \geq \gamma.$$

The length parameter $N \geq 1$ caters for scenarios where a single sample from $\mathcal{C}$ or $\tilde{\mathcal{C}}$ may not be sufficient for embedding. E.g. in Section 5.3 where we rely on rejection-sampling, the approach forbids a sample of size $\geq \omega(\log \lambda)$ by nature, and we require $N > 1$ tails to collectively embed a PKE ciphertext.

The purpose of ChkT is to check and record the suitability of tails for embedding.¹⁴ In Sections 5.3 and 5.4, we work over channels $\mathcal{C}$ which are embedding friendly everywhere and our NSE constructions (Figs. 3 and 4) simply have ChkT returning $\perp$. The algorithm ChkT will become useful in our concrete implementation in Section 6.

5.2 CCA PKS for 0-memoryless channels with NSE

Construction. In Fig. 2 we construct a PKS Π for any pair of 0-memoryless channels $\mathcal{C} = (\mathcal{C}_i)_{i \in \mathbb{N}}$ and $\tilde{\mathcal{C}} = (\tilde{\mathcal{C}}_i)_{i \in \mathbb{N}}$ with a common alphabet $\Sigma = \mathbb{H} \times \mathbb{T}$ and which admits an NSE with length parameter N , head message alphabet $\mathcal{M}_H$ and tail message alphabet $\mathcal{M}_T$. We further need the following properties or building blocks:

- $(\mathcal{M}_H, \oplus)$ is a finite group with subtraction denoted as $\ominus$.
- ECC, an error-correcting code with a sufficiently large implicitly defined message space and with codeword space $\mathcal{M}_H^N$.
- PKE, a public-key encryption scheme with message space $\mathcal{K}$ and ciphertext space $\mathcal{M}_T^N$.
- PRF, a pseudorandom function $\{0, 1\}^\lambda \times [N] \rightarrow \mathcal{M}_H$.
- G_1, G_2 , hash functions which output sufficiently long bit strings to be modelled as random oracles in the security proof.

In a nutshell, the construction combines a CPA-secure symmetric-key stegosystem and a CPA-secure public-key stegosystem using the FO transform [FO13] in a non-black-box way. The CCA-security of the construction relies on passing the randomness derived in the FO transform to EmbT for tail embedding.

In more detail, the stegoencoder (1) draws N samples from the channel $\mathcal{C}$ consisting of N head-tail tuples, (2) encodes the hidden message and the tails as cw using ECC, (3) use PRF as a symmetric-key encryption scheme to encrypt the ECC codeword as $\widetilde{\text{cw}}$, with secret key derived from G_1 , (4) use EmbH to embed the encrypted codeword $\widetilde{\text{cw}}$ into the heads, (5) use G_2 to derive PKE encryption randomness and tail embedding randomness from the embedded heads, (6) use PKE to encrypt the PRF secret key with the derived randomness, and (7) use EmbT to embed the PKE ciphertext into the tails using the derived randomness. Correspondingly, the stegodecoder first performs a trial decryption, and only releases the decryption result if re-encryption recovers the given stegotext.

Correctness. To show correctness of Π in Fig. 2, first we observe that the pair of subroutines $(\text{EncEK}, \text{DecEK})$ can be viewed as an “outer PKE” which uses PKE. The following lemma says that it is correct.

Lemma 5. *Suppose PKE is correct and NSE has correct tail embedding. Let $(\text{EncEK}, \text{DecEK})$ be as defined in Fig. 2. Then, for all $(\text{pk}, \text{sk}) \in \text{PKE.KGen}$, $\text{ek} \in \mathcal{K}$, $\text{tails} \in \mathbb{T}^N$ and $\text{aux} \in \text{ChkT}(\text{tails})$,*

$$\Pr[\text{DecEK}(\text{sk}, \text{EncEK}(\text{pk}, \text{ek}, \text{tails}, \text{aux})) = \text{ek}] \geq 1 - \text{negl}(\lambda).$$

Theorem 1 below shows that Π is correct as a PKS. It relies on correctness of ECC, NSE and $(\text{EncEK}, \text{DecEK})$.

Theorem 1 (Correctness). *Suppose PKE is correct, ECC is correct and NSE has correct head embedding and correct tail embedding. Then the PKS construction Π in Fig. 2 is $(\mathcal{C}, \tilde{\mathcal{C}})$ -correct.*

The proofs of Lemma 5 and Theorem 1 are standard. We defer them to Appendices B.1 and B.2.

¹⁴ For example, if for certain $i \in \mathbb{N}$ the entropy of tail_i is too low for embedding any message, this information can be recorded in aux to aid the decoder.

$\text{Enc}^{\mathcal{C}_{\geq \text{hist}}, \tilde{\mathcal{C}}_{\geq \text{hist}}}(\text{pk}, \text{msg}, \text{hist})$ <pre> 1 : for $i \in [N]$ do 2 : $c_i \leftarrow \mathcal{C}_{\text{hist} \ c_0 \ \dots \ c_{i-1}}$ 3 : parse $(\text{head}_i, \text{tail}_i) \leftarrow c_i$ 4 : $\text{tails} := (\text{tail}_i)_{i \in [N]}$; $\text{aux} \leftarrow \text{NSE.ChkT}(\text{tails})$ 5 : $\text{cw} = (\text{cw}_i)_{i \in [N]} \leftarrow \text{ECC.Enc}(\text{msg}, \text{tails}, \text{aux})$ 6 : $\text{ek} \leftarrow \mathcal{K}$ // ephemeral key 7 : $\text{sk}_{\text{PRF}} \leftarrow \text{G}_1(\text{pk}, \text{ek})$ 8 : // one-time-pad cw_i and embed to head_i: 9 : for $i \in [N]$ do 10 : $\widetilde{\text{cw}}_i \leftarrow \text{PRF}(\text{sk}_{\text{PRF}}, i) \oplus \text{cw}_i$ 11 : $\widetilde{\text{head}}_i \leftarrow \text{NSE.EmbH}^{\tilde{\mathcal{C}}_{\geq \text{hist}}}(\text{hist} + i, \text{head}_i, \widetilde{\text{cw}}_i)$ 12 : $\widetilde{\text{heads}} := (\widetilde{\text{head}}_i)_{i \in [N]}$ 13 : // encrypt and embed ek to tails: 14 : $r \leftarrow \text{G}_2(\text{pk}, \text{ek}, \widetilde{\text{heads}})$ 15 : $\widetilde{\text{tails}} \leftarrow \text{EncEK}(\text{pk}, \text{ek}, \text{tails}, \text{aux}; r)$ 16 : $\tilde{c}_i := (\widetilde{\text{head}}_i, \widetilde{\text{tail}}_i)$ 17 : return $\text{stxt} := (\tilde{c}_i)_{i \in [N]}$ </pre> $\text{EncEK}(\text{pk}, \text{ek}, \text{tails}, \text{aux}; r)$ <pre> 1 : parse $(r_1, (r_{2,i})_{i \in [N]}) \leftarrow r$ 2 : $\text{ctxt} \leftarrow \text{PKE.Enc}(\text{pk}, \text{ek}; r_1)$ 3 : parse $(\text{ctxt}_i)_{i \in [N]} \leftarrow \text{ctxt}$ 4 : for $i \in [N]$ do 5 : $\widetilde{\text{tail}}_i \leftarrow \text{NSE.EmbT}(\text{hist} + i, \text{tail}_i, \text{ctxt}_i, \text{aux}; r_{2,i})$ 6 : $\widetilde{\text{tails}} := (\widetilde{\text{tail}}_i)_{i \in [N]}$ 7 : return $\widetilde{\text{tails}}$ </pre>	$\text{KGen}(1^\lambda)$ <pre> 1 : return $(\text{pk}, \text{sk}) \leftarrow \text{PKE.KGen}(1^\lambda)$ </pre> $\text{Dec}(\text{sk}, \text{stxt})$ <pre> 1 : parse $(\widetilde{\text{head}}_i, \widetilde{\text{tail}}_i)_{i \in [N]} \leftarrow \text{stxt}$ 2 : $\widetilde{\text{heads}} := (\widetilde{\text{head}}_i)_{i \in [N]}$ 3 : $\widetilde{\text{tails}} := (\widetilde{\text{tail}}_i)_{i \in [N]}$ 4 : // trial decryption: 5 : $\text{ek} \leftarrow \text{DecEK}(\text{sk}, \widetilde{\text{tails}})$ 6 : $\text{sk}_{\text{PRF}} \leftarrow \text{G}_1(\text{pk}, \text{ek})$ 7 : for $i \in [N]$ do 8 : $\widetilde{\text{cw}}_i \leftarrow \text{NSE.ExtH}(\widetilde{\text{head}}_i)$ 9 : $\text{cw}_i \leftarrow \widetilde{\text{cw}}_i \ominus \text{PRF}(\text{sk}_{\text{PRF}}, i)$ 10 : $\text{cw} := (\text{cw}_i)_{i \in [N]}$ 11 : $(\text{msg}, \text{tails}, \text{aux}) \leftarrow \text{ECC.Dec}(\text{cw})$ 12 : // re-encryption check: 13 : $r \leftarrow \text{G}_2(\text{pk}, \text{ek}, \widetilde{\text{heads}})$ 14 : assert $\text{EncEK}(\text{pk}, \text{ek}, \text{tails}, \text{aux}; r) = \widetilde{\text{tails}}$ 15 : return msg </pre> $\text{DecEK}(\text{sk}, \widetilde{\text{tails}})$ <pre> 1 : parse $(\widetilde{\text{tail}}_i)_{i \in [N]} \leftarrow \widetilde{\text{tails}}$ 2 : for $i \in [N]$ do 3 : $\text{ctxt}_i \leftarrow \text{NSE.ExtT}(\widetilde{\text{tail}}_i)$ 4 : $\text{ctxt} := (\text{ctxt}_i)_{i \in [N]}$ 5 : $\text{ek} \leftarrow \text{PKE.Dec}(\text{sk}, \text{ctxt})$ 6 : return ek </pre>
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Fig. 2. PKS construction Π based on ECC, PKE, PRF, G_1 , G_2 and NSE.

Security. In the classic FO transform [FO13], its underlying PKE is required to satisfy a well-spreadness property. In our case, this is replaced by the well-spreadness of NSE. Lemma 6 says that if NSE is well-spread, then so is EncEK.

Lemma 6. *Suppose NSE has tail γ -spreadness for some γ satisfying $\gamma \cdot N \geq \omega(\log \lambda)$. Let PKE be with ciphertext space $\mathcal{M}_T^N$ and EncEK be as defined in Fig. 2. Then, for any $(\text{pk}, \text{sk}) \in \text{PKE.KGen}$, $\text{ek} \in \mathcal{K}$, $\text{tails} \in \mathbb{T}^N$, $\text{aux} \in \text{ChkT}(\text{tails})$,*

$$H_\infty(\text{EncEK}(\text{pk}, \text{ek}, \text{tails}, \text{aux})) \geq \omega(\log \lambda).$$

Proof. By design of EncEK, it returns the N outputs from EmbT, each using independent randomness. The claim then follows directly from that min-entropy is multiplicative. $\square$

Theorem 2 below states the CCA-security of Π as a PKS. Its proof is somewhat similar to the original proof of the FO transform. A major difference is that we rely on the indistinguishability property of NSE (Definition 9), which allows to argue the indistinguishability between a stegotext and a coverttext.

Theorem 2 (CCA Security). *Suppose the following hold:*

- PKE is correct and IND-\$-CPA secure,
- PRF is secure,
- NSE has tail embedding correctness, tail γ -spreadness where $\gamma \cdot N \geq \omega(\log \lambda)$, and $(\mathcal{C}, \tilde{\mathcal{C}}, \epsilon)$ indistinguishability for some $\epsilon \leq \text{negl}(\lambda)$, and
- the message space $\mathcal{K}$ of PKE satisfies $|\mathcal{K}| \geq \lambda^{\omega(1)} \cdot \text{poly}(\lambda)$.

Then, Π in Fig. 2 is a $(\mathcal{C}, \tilde{\mathcal{C}})$ -IND-CCA secure PKS in the random oracle model.

Proof. We define the following sequence of hybrid experiments:

Hyb₀: The real $(\mathcal{C}, \tilde{\mathcal{C}})$ -IND-CCA security experiment for Π with bit $b = 0$. We model G_1 and G_2 by random oracles. The full decoding oracle DecO is recalled in Fig. 6 in Appendix B.3.

Hyb₁ (Oracle answers without sk^*): Almost identical to Hyb₀, except that we swap DecO and G_1 to DecO[†] defined in Fig. 6. Note that DecO[†] no longer uses the PKE secret key sk^* to answer the queries.

We have $\text{Hyb}_0 \approx_s \text{Hyb}_1$ by Lemma 13.

Hyb₂ (G_2 rejects answering to ek^*): Almost identical to Hyb₁, except that the experiment aborts if the adversary $\mathcal{A}$ queries G_2 on $(\text{pk}^*, \text{ek}^*, \widetilde{\text{heads}}, \widetilde{\text{tails}}, \text{aux})$ for any $\text{aux}, \widetilde{\text{heads}}, \widetilde{\text{tails}}$, where pk^*, ek^* are the public key and ephemeral key used for generating the challenge stegotext.

We have $\text{Hyb}_1 \approx_c \text{Hyb}_2$ under the IND-CPA security of PKE by Lemma 14.

Hyb₃ (Random PKE ciphertext): Almost identical to Hyb₂, except that in $\Pi.\text{Enc}$, we swap $\text{ctxt} \leftarrow \text{PKE}.\text{Enc}(\text{pk}^*, \text{ek}^*; r_1)$ to a uniformly random vector $\text{ctxt} \leftarrow \mathcal{M}_T^N$.

We have $\text{Hyb}_2 \approx_c \text{Hyb}_3$ under the IND-\$-CPA security of PKE by Lemma 15.

Hyb₄ (G_1 rejects answering to ek^*): Almost identical to Hyb₃, except that G_1 returns $\perp$ if being queried on $(\text{pk}^*, \text{ek}^*)$ after the challenge stegotext is defined, where pk^*, ek^* are the public key and ephemeral key used for generating the challenge stegotext.

We have $\text{Hyb}_3 \approx_s \text{Hyb}_4$ by Lemma 16.

Hyb₅ (Random PRF-padded value): Almost identical to Hyb₄, except that when generating the challenge stegotext, in $\Pi.\text{Enc}$, for each $i \in [N]$, we swap $\widetilde{\text{cw}}_i \leftarrow \text{PRF}(\text{sk}_{\text{PRF}}^*; i) \oplus \text{cw}_i$ to uniformly random $\widetilde{\text{cw}}_i \leftarrow \mathcal{M}_H$.

We have $\text{Hyb}_4 \approx_c \text{Hyb}_5$ under the security of PRF by Lemma 17.

Hyb₆: The ideal $(\mathcal{C}, \tilde{\mathcal{C}})$ -IND-CCA security experiment for Π with bit $b = 1$.

We have $\text{Hyb}_5 \approx_s \text{Hyb}_6$ under the $(\mathcal{C}, \tilde{\mathcal{C}}, \epsilon)$ -indistinguishability of NSE by Lemma 18.

Lemmas 13 to 18 are provided in Appendix B.3. Putting the above together completes the proof. $\square$

5.3 NSE via rejection sampling for partially sampleable channels

We give a generic instantiation of NSE (Definition 9) based on rejection sampling for a class of channels which we call “partially sampleable channels”. A 0-memoryless channel $\mathcal{C} = (\mathcal{C}_i)_{i \in \mathbb{N}}$ is partially sampleable if part of the output of $\mathcal{C}_i$ can be independently resampled with an efficient algorithm, i.e. with explicit randomness. Formally:

Definition 10 (Partially sampleable channels (PSC)). *Two 0-memoryless channels $\mathcal{C} = (\mathcal{C}_i)_{i \in \mathbb{N}}$ and $\tilde{\mathcal{C}} = (\tilde{\mathcal{C}}_i)_{i \in \mathbb{N}}$ with a common alphabet $\Sigma = \mathbb{H} \times \mathbb{T}$ is said to be partially sampleable if there exists a PPT algorithm SampT such that the following distribution is identical to $\tilde{\mathcal{C}}_i$ for each $i \in \mathbb{N}$:*

$$\left\{ \left(\widetilde{\text{head}}_i, \widetilde{\text{tail}}'_i \right) \in \Sigma \mid \begin{array}{l} (\text{head}_i, \text{tail}_i) \leftarrow \mathcal{C}_i; \quad (\widetilde{\text{head}}_i, \widetilde{\text{tail}}_i) \leftarrow \mathcal{C}_i \\ \widetilde{\text{tail}}'_i \leftarrow \text{SampT}(i, \text{tail}_i) \end{array} \right\}.$$

In particular, SampT could be given explicit randomness (from $\{0, 1\}^\lambda$ without loss of generality) as input.

W.l.o.g. we write $\tilde{\mathcal{C}} = \tilde{\mathcal{C}}_{\mathbb{H}} \times \tilde{\mathcal{C}}_{\mathbb{T}}$, where $\tilde{\mathcal{C}}_{\mathbb{H}} = (\tilde{\mathcal{C}}_{\mathbb{H}, i})_{i \in \mathbb{N}}$ and $\tilde{\mathcal{C}}_{\mathbb{T}} = (\tilde{\mathcal{C}}_{\mathbb{T}, i})_{i \in \mathbb{N}}$ are the partial channels of $\tilde{\mathcal{C}}$ corresponding to $\widetilde{\text{head}}_i$ ’s and $\widetilde{\text{tail}}_i$ ’s.

$\text{EmbH}^{\tilde{\mathcal{C}} \geq i}(i, \text{head}_i, \text{msg}_{h_i})$	$\text{EmbT}(i, \text{tail}_i, \text{msg}_{t_i}, \text{aux}; r)$
for counter $\in [\lambda]$ do	for counter $\in [\lambda]$ do
$(\widetilde{\text{head}}_i, \widetilde{\text{tail}}_i) \leftarrow \tilde{\mathcal{C}}_i$	$\widetilde{\text{tail}}_i \leftarrow \text{SampT}(i, \text{tail}_i)$
if $H_1(\widetilde{\text{head}}_i) = \text{msg}_{h_i}$ then return $\widetilde{\text{head}}_i$	if $H_2(\widetilde{\text{tail}}_i) = \text{msg}_{t_i}$ then return $\widetilde{\text{tail}}_i$
return $\perp$	return $\perp$
$\text{ExtH}(\widetilde{\text{head}}_i)$	$\text{ExtT}(\widetilde{\text{tail}}_i)$
return $\text{msg}_{h_i} := H_1(\widetilde{\text{head}}_i)$	return $\text{msg}_{t_i} := H_2(\widetilde{\text{tail}}_i)$

Fig. 3. $\text{NSE}_{\text{RejSamp}}$ for PSC $(\mathcal{C}, \tilde{\mathcal{C}})$ via rejection-sampling. $\text{ChkT}(\text{tails})$ returns $\text{aux} = \perp$.

We think of the heads of $\mathcal{C}, \tilde{\mathcal{C}}$ as some uncontrollable distributions, e.g. which have semantic meaning, and the tails as some efficiently sampleable distributions. In the context of steganography, even when no explicit efficient algorithms for sampling from $\tilde{\mathcal{C}}_i$ is known, it can still be reasonable that part of $\tilde{\mathcal{C}}_i$ (its tails) can be efficiently sampled, especially if one is given a sample from a related channel $\mathcal{C}_i$. This captures, for example, raw digital photographs where the ISO noise follows a more or less well-understood distribution, Internet packets where some entries contain random nonces, audio tracks with background noise, etc.

Remark 3 (Relaxation to statistical or computational closeness.). In Definition 10 we require $\tilde{\mathcal{C}}_i$ and the distribution obtained by replacing $\widetilde{\text{tail}}_i$ with a sample $\widetilde{\text{tail}}'_i$ from SampT to be identical. This can be relaxed to requiring only statistical or computational closeness, with Lemma 8 and Theorem 3 in the following adapted accordingly. Specifically, with such a relaxed PSC definition, the head and tail of $\tilde{\mathcal{C}}$ are no longer guaranteed to be independent; instead, one deduces that $\tilde{\mathcal{C}} \approx \tilde{\mathcal{C}}'_H \times \tilde{\mathcal{C}}'_T$ (where $\approx$ denotes the corresponding statistical or computational closeness) for two independent channels $\tilde{\mathcal{C}}'_H$ and $\tilde{\mathcal{C}}'_T$, defined by $\tilde{\mathcal{C}}'_{H,i} = \{\widetilde{\text{head}}_i \mid (\widetilde{\text{head}}_i, \widetilde{\text{tail}}_i) \leftarrow \mathcal{C}_i\}$ and $\tilde{\mathcal{C}}'_{T,i} = \{\widetilde{\text{tail}}_i \mid (\widetilde{\text{head}}_i, \widetilde{\text{tail}}_i) \leftarrow \mathcal{C}_i; \widetilde{\text{tail}}'_i \leftarrow \text{SampT}(i, \text{tail}_i)\}$ respectively. The adapted Lemma 8 and Theorem 3 would require sufficiently large min-entropy for such intermediate channels $\tilde{\mathcal{C}}'_H$ and $\tilde{\mathcal{C}}'_T$.

Construction. Let $(\mathcal{C}, \tilde{\mathcal{C}})$ be any PSC (Definition 10). Let $H_1 : \mathbb{H} \rightarrow \mathcal{M}_H, H_2 : \mathbb{T} \rightarrow \mathcal{M}_T$ be two strongly-2-universal hash functions. We give an NSE construction $\text{NSE}_{\text{RejSamp}}$ in Fig. 3 for $(\mathcal{C}, \tilde{\mathcal{C}})$. At a high level, it works in a similar way as the stegosystem of [BC05] based on rejection sampling; we adapt it to the context of PSC which rejection-samples each of the stegotext components $\widetilde{\text{head}}_i, \widetilde{\text{tail}}_i$.

The following lemma from [BC05] gives a bound on the statistical distance between a sample from a channel and one from rejection sampling.

Lemma 7 (Adapted from [BC05, Proposition 7 (eprint)]). *Let $\mathcal{C}$ be a channel with alphabet Σ and min-entropy $H_\infty(\mathcal{C})$. Let $H : \Sigma \rightarrow \mathcal{M}$ be a strongly-2-universal hash function. For any fixed $m \in \mathcal{M}$, consider the following two distributions of c :*

1. *Sample $c \leftarrow \mathcal{C}$.*
2. *Do the following for λ times: Sample $c \leftarrow \mathcal{C}$, check whether $H(c) = m$. If yes then break and return c ; if no then repeat. If nothing is returned after λ trials, return $c := \perp$.*

The statistical distance between the above two distributions of c is upper-bounded by $2^{\log |\mathcal{M}| - H_\infty(\mathcal{C})} + \alpha^\lambda$ for some constant $\alpha < 1$.

The next lemma shows that, for any PSC $(\mathcal{C}, \tilde{\mathcal{C}})$ which has sufficient entropy, $\text{NSE}_{\text{RejSamp}}$ achieves the desired properties of an NSE stated in Definition 9. In brief, correctness follows from H_1, H_2 being strongly-2-universal hashes and thus they map samples from $\tilde{\mathcal{C}}$ to uniform outputs if $\tilde{\mathcal{C}}$ has high enough entropy, and indistinguishability relies on Lemma 7. Its proof is given in Appendix B.4.

Lemma 8. *The NSE construction $\text{NSE}_{\text{RejSamp}}$ in Fig. 3 is PPT. Suppose $\tilde{\mathcal{C}}$ has:*

- *alphabet $\mathbb{H} \times \mathbb{T}$ where $|\mathbb{H}| \leq 2^{\gamma_H \cdot \log \lambda}$ and $|\mathbb{T}| \leq 2^{\gamma_T \cdot \log \lambda}$ for some constants $\gamma_H, \gamma_T > 0$,*
- *sufficiently large head min-entropy $H_\infty(\tilde{\mathcal{C}}_{\mathbb{H}}) \geq |\mathcal{M}_H| + \omega(\log \lambda)$,*
- *sufficiently large tail min-entropy $H_\infty(\tilde{\mathcal{C}}_{\mathbb{T}}) \geq |\mathcal{M}_T| + \omega(\log \lambda)$.*

Then, $\text{NSE}_{\text{RejSamp}}$ has correct head embedding, correct tail embedding, $(\mathcal{C}, \tilde{\mathcal{C}}, \epsilon)$ -indistinguishability, and tail γ -spreadness, for some $\epsilon \leq \text{negl}(\lambda)$ and $\gamma = H_\infty(\tilde{\mathcal{C}}_{\mathbb{T}})$.

Using Theorems 1 and 2 and Lemma 8, we obtain Theorem 3, which circumvents the impossibility of [BL18] – There exists an IND-CCA PKS for the subclass PSC of the class of 0-memoryless channels.

Theorem 3 (CCA PKS for 0-memoryless PSC). *Let $(\mathcal{C}, \tilde{\mathcal{C}})$ be any PSC (Definition 10, possibly $\mathcal{C} = \tilde{\mathcal{C}}$) satisfying the conditions stated in Lemma 8. Let Π_{RejSamp} be the PKS Π in Fig. 2 instantiated with:*

- *a correct ECC,*
- *a secure PRF,*
- *a correct and IND- $\mathcal{S}$ -CPA secure PKE with a message space of size super-polynomial in λ , and*
- *the $\text{NSE}_{\text{RejSamp}}$ in Fig. 3.*

Then, Π_{RejSamp} is a correct and $(\mathcal{C}, \tilde{\mathcal{C}})$ -IND-CCA-secure PKS in the random oracle model.

Rate. Recall that the message space is $\mathcal{M} = \mathcal{M}_H^N$, and the stegotext space is $\mathcal{S} = \mathbb{H}^N \times \mathbb{T}^N$, hence the rate of Π_{RejSamp} in Theorem 3 is $\log |\mathcal{M}| / \log |\mathcal{S}| = |\mathcal{M}_H| / (\log |\mathbb{H}| + \log |\mathbb{T}|) \leq |\mathcal{M}_H| / H_\infty(\tilde{\mathcal{C}}_{\mathbb{H}}) \leq O(\log \lambda) / H_\infty(\tilde{\mathcal{C}}_{\mathbb{H}})$, where the second inequality is due to the constraints from Lemma 8. In particular, the rate diminishes as the entropy $H_\infty(\tilde{\mathcal{C}}_{\mathbb{H}})$ of $\tilde{\mathcal{C}}$ increases.

5.4 NSE via preimage-sampling for Gaussian channels

Our second NSE works over channels following Gaussian distributions, it relies on Gaussian preimage-sampling technique from lattice-based cryptography.

Definition 11 (Twin-Gaussian channels (TGC)). *Two 0-memoryless channels $\mathcal{C} = (\mathcal{C}_i)_{i \in \mathbb{N}}$ and $\tilde{\mathcal{C}} = (\tilde{\mathcal{C}}_i)_{i \in \mathbb{N}}$ with a common alphabet $\Sigma = \mathbb{H} \times \mathbb{T}$ is called (m_H, m_T, D) -twin-Gaussian, if $\mathbb{H} = \mathbb{Z}^{m_H}$, $\mathbb{T} = \mathbb{Z}^{m_T}$, and $(\mathcal{C}, \tilde{\mathcal{C}})$ is a pair of Gaussian distributions with description $D = (\Theta_{\mathbf{v}_j}, \Phi_{\mathbf{v}_j}, \mathbf{v}_j)_{j \in \{h, t\} \times \mathbb{N}}$, such that*

$$\begin{aligned} \mathcal{C}_i &\equiv \mathcal{D}_{\mathbb{Z}^{m_H}, \sqrt{\Theta_{\mathbf{v}_{h,i}}}, \mathbf{v}_{h,i}} \times \mathcal{D}_{\mathbb{Z}^{m_T}, \sqrt{\Theta_{\mathbf{v}_{t,i}}}, \mathbf{v}_{t,i}}, \\ \tilde{\mathcal{C}}_i &\equiv \mathcal{D}_{\mathbb{Z}^{m_H}, \sqrt{\Phi_{\mathbf{v}_{h,i}}}, \mathbf{v}_{h,i}} \times \mathcal{D}_{\mathbb{Z}^{m_T}, \sqrt{\Phi_{\mathbf{v}_{t,i}}}, \mathbf{v}_{t,i}}. \end{aligned}$$

In words, for any index i , $\mathcal{C}$ and $\tilde{\mathcal{C}}$ are both a tuple of two independent Gaussians, they have the same center $(\mathbf{v}_{h,i}, \mathbf{v}_{t,i})$ and whose width is dependent on $(\mathbf{v}_{h,i}, \mathbf{v}_{t,i})$. The difference between the two channels is that their width parameter can differ. For concreteness, think of $\mathcal{C}, \tilde{\mathcal{C}}$ as channels both transmitting the values $(\mathbf{v}_{h,i}, \mathbf{v}_{t,i})$ but injected with different magnitudes of Gaussian noises. Real-life examples of such channels are, for example, photos taken by a camera with different ISO settings, or audio recorded with different gains.

Construction. Let $(\mathcal{C}, \tilde{\mathcal{C}})$ be a pair of (m_H, m_T, D) -twin-Gaussian channels (Definition 11), and let the message alphabets be $\mathcal{M}_H = \mathbb{Z}_q^{n_H}$ and $\mathcal{M}_T = \mathbb{Z}_q^{n_T}$, such that it holds $m_H = n_H \lceil \log_\beta q \rceil$ and $m_T = n_T \lceil \log_\beta q \rceil$ for some integers $q \geq \beta \geq 2$ and q is prime. Let SampPre be the PPT algorithm given by Lemma 4. In Fig. 4 we give an NSE construction $\text{NSE}_{\text{GaussSamp}}$ for $(\mathcal{C}, \tilde{\mathcal{C}})$.

Lemma 9. *The NSE construction $\text{NSE}_{\text{GaussSamp}}$ in Fig. 3 is PPT, it has correct head embedding and correct tail embedding.*

$\text{EmbH}^{\tilde{\mathcal{C}} \geq i}(i, \text{head}_i = \mathbf{h}_i \in \mathbb{Z}^{m_H}, \text{msg}_i \in \mathbb{Z}_q^{n_H})$	$\text{ExtH}(\widetilde{\text{head}_i} = \tilde{\mathbf{h}}_i)$
$\tilde{\mathbf{h}}_i \leftarrow \text{Emb}(\mathbf{h}_i, \text{msg}_i, n_H, \beta)$	$\text{msg}_i \leftarrow \text{Ext}(\tilde{\mathbf{h}}_i, n_H, \beta)$
return $\tilde{\mathbf{h}}_i$	return msg_i
$\text{EmbT}(i, \text{tail}_i = \mathbf{t}_i \in \mathbb{Z}^{m_T}, \text{msgt}_i \in \mathbb{Z}_q^{n_T}, \text{aux}; r)$	$\text{ExtT}(\widetilde{\text{tail}_i} = \tilde{\mathbf{t}}_i)$
$\tilde{\mathbf{t}}_i \leftarrow \text{Emb}(\mathbf{t}_i, \text{msgt}_i, n_T, \beta; r)$	$\text{msgt}_i \leftarrow \text{Ext}(\tilde{\mathbf{t}}_i, n_T, \beta)$
return $\tilde{\mathbf{t}}_i$	return msgt_i
$\text{Emb}(\mathbf{x}, \mathbf{y}, n, \beta; r)$	$\text{Ext}(\tilde{\mathbf{x}}, n, \beta)$
$\Delta_{\mathbf{x}} := \Phi_{\mathbf{x}} - \Theta_{\mathbf{x}} \quad // \text{ Gaussian param. according to } \mathbf{x}$	$\mathbf{y} := \mathbf{G}_{n, \beta} \cdot \tilde{\mathbf{x}} \bmod q$
return $\tilde{\mathbf{x}} \leftarrow \text{SampPre}(n, \beta, \mathbf{y}, \sqrt{\Delta_{\mathbf{x}}}, \mathbf{x}; r)$	return $\mathbf{y}$

Fig. 4. $\text{NSE}_{\text{GaussSamp}}$ for TGC $(\mathcal{C}, \tilde{\mathcal{C}})$ via Gaussian preimage-sampling. $\text{ChkT}(\text{tails})$ returns $\text{aux} = \perp$.

Proof. $\text{NSE}_{\text{GaussSamp}}$ is PPT since SampPre is and Ext is deterministic. Its head embedding and tail embedding correctness is by Lemma 4 Item 1. $\square$

Theorem 4. Let $(\mathcal{C}, \tilde{\mathcal{C}})$ be a pair of (m_H, m_T, D) -TGC with description $D = (\Theta_{\mathbf{v}_j}, \Phi_{\mathbf{v}_j}, \mathbf{v}_j)_{j \in \{h, t\} \times \mathbb{N}}$, such that $n_H = m_H \lceil \log_{\beta} q \rceil$, $n_T = m_T \lceil \log_{\beta} q \rceil$ for some $n_H, n_T \in \mathbb{N}$. Suppose the following hold for any $(n, m) \in \{(n_H, m_H), (n_T, m_T)\}$, $\mathbf{v} \in \mathbb{R}^m$, and $\Delta_{\mathbf{v}} := \Phi_{\mathbf{v}} - \Theta_{\mathbf{v}}$:

- Over the randomness of $\mathbf{x} \leftarrow \mathcal{D}_{\mathbb{Z}^m, \sqrt{\Theta_{\mathbf{v}}}, \mathbf{v}}$, it holds that
 - $\Pr[\sqrt{\Delta_{\mathbf{x}}} \geq 2 \cdot \eta_{\epsilon}(A_q^{\perp}(\mathbf{G}_{n, \beta}))] \geq 1 - \text{negl}(\lambda)$ for some $\epsilon \leq \text{negl}(\lambda)$,¹⁵
 - $\Pr[\rho_{\sqrt{\Delta_{\mathbf{x}}}}(\mathbf{z}) \in [1 - \beta_0, 1 + \beta_1] \cdot \rho_{\sqrt{\Delta_{\mathbf{v}}}}(\mathbf{z}) \quad \forall \mathbf{z} \in \mathbb{Z}^m] \geq 1 - \text{negl}(\lambda)$ for some $\beta_0, \beta_1 \in \mathbb{R}_{\geq 0}$.
- $\Delta_{\mathbf{v}} > \eta_{\epsilon}(\mathbb{Z}^m)$ and $\sqrt{\Theta_{\mathbf{v}}^{-1}} + \Delta_{\mathbf{v}}^{-1} > \eta_{\epsilon}(\mathbb{Z}^m)$ for some $\epsilon \leq \text{negl}(\lambda)$,

Then, the NSE construction $\text{NSE}_{\text{GaussSamp}}$ in Fig. 4 has $(\mathcal{C}, \tilde{\mathcal{C}}, \epsilon)$ -indistinguishability where

$$\epsilon \leq 2N \cdot \left(\frac{\beta_0 + \beta_1}{2(1 - \beta_0)} + \text{negl}(\lambda) \right),$$

and it has tail γ -spreadness where $\gamma \geq m - O(1)$.

Proof. The tail spreadness condition follows directly from Lemma 4 Item 2 and Lemma 2. In the rest we bound the statistical distance between the two distributions in Eq. (1).

The definition of TGC implies that, for any fixed $\text{msg}_i \in \mathbb{Z}_q^{n_H}$ and $\text{msgt}_i \in \mathbb{Z}_q^{n_T}$, the distributions of $\text{head}_i = \tilde{\mathbf{h}}_i$ and $\text{tail}_i = \tilde{\mathbf{t}}_i$ are independent of each other and for any index $i \in [N]$. Therefore, it suffices to individually bound the statistical distance of head_i resp. tail_i for each i , i.e. that between

$$\begin{aligned} & \left\{ \tilde{\mathbf{h}}_i \mid \begin{array}{l} (\mathbf{h}_i, \mathbf{t}_i) \leftarrow \mathcal{C}; \text{msg}_i \leftarrow \mathbb{Z}_q^{n_H} \\ \tilde{\mathbf{h}}_i \leftarrow \text{EmbH}^{\tilde{\mathcal{C}}_i}(i, \mathbf{h}_i, \text{msg}_i) \end{array} \right\} & \text{and} & \left\{ \tilde{\mathbf{h}}_i \mid (\tilde{\mathbf{h}}_i, \tilde{\mathbf{t}}_i) \leftarrow \tilde{\mathcal{C}} \right\}, \\ \text{resp.} & \left\{ \tilde{\mathbf{t}}_i \mid \begin{array}{l} (\mathbf{h}_i, \mathbf{t}_i) \leftarrow \mathcal{C}; \text{msgt}_i \leftarrow \mathbb{Z}_q^{n_T} \\ \tilde{\mathbf{t}}_i \leftarrow \text{EmbT}(i, \mathbf{t}_i, \text{msgt}_i, \perp) \end{array} \right\} & \text{and} & \left\{ \tilde{\mathbf{t}}_i \mid (\tilde{\mathbf{h}}_i, \tilde{\mathbf{t}}_i) \leftarrow \tilde{\mathcal{C}} \right\}, \end{aligned} \quad (2)$$

then the claim follows from repeating the same argument over all $i \in [N]$.

Now observe that in Fig. 4, both EmbH and EmbT make a call to Emb , the only difference is the input parameters. Our argument for $\text{head}_i = \tilde{\mathbf{h}}_i$ and $\text{tail}_i = \tilde{\mathbf{t}}_i$ will be the same, where we go through a sequence of hybrids gradually changing the computation of Emb .

¹⁵ The multiplicative factor of 2 in the lower bound is only for arguing tail spreadness using Lemma 2. This may be relaxed given a tighter Gaussian entropy bound.

Below, fix any $i \in [N]$ and consider either $(\mathbf{x} = \mathbf{h}_i, \mathbf{y} = \text{msg}h_i, \mathbf{v} = \mathbf{v}_{h,i}, n = n_H, m = m_H)$ or $(\mathbf{x} = \mathbf{t}_i, \mathbf{y} = \text{msg}t_i, \mathbf{v} = \mathbf{v}_{t,i}, n = n_T, m = m_T)$.

Hyb₀: The LHS distribution in Eq. (2). Here, the underlying algorithm Emb samples and returns

$$\tilde{\mathbf{x}} \leftarrow \$ \text{SampPre}(n, \beta, \mathbf{y}, \sqrt{\Delta_{\mathbf{x}}}, \mathbf{x}),$$

where the Gaussian parameter $\Delta_{\mathbf{x}} = \Phi_{\mathbf{x}} - \Theta_{\mathbf{x}}$ is determined by the channel descriptions and the input $\mathbf{x}$.

Hyb₁: Almost identical to Hyb₀, except that Emb returns

$$\tilde{\mathbf{x}} \leftarrow \$ \mathcal{D}_{A_q^{\mathbf{y}}(\mathbf{G}_{n,\beta}), \sqrt{\Delta_{\mathbf{x}}}, \mathbf{x}},$$

i.e. inefficiently sampling discrete Gaussian subject to $\mathbf{G}_{n,\beta} \cdot \tilde{\mathbf{x}} = \mathbf{y} \bmod q$.

Hyb₂: Almost identical to Hyb₁, except that Emb returns

$$\tilde{\mathbf{x}} \leftarrow \$ \mathcal{D}_{\mathbb{Z}^m, \sqrt{\Delta_{\mathbf{x}}}, \mathbf{x}},$$

i.e. discrete Gaussian over $\mathbb{Z}^m$, with the same parameter $\sqrt{\Delta_{\mathbf{x}}}$ and center $\mathbf{x}$ but without constraint from $\mathbf{y}$.

Hyb₃: Almost identical to Hyb₂, except that Emb returns

$$\tilde{\mathbf{x}} \leftarrow \$ \mathcal{D}_{\mathbb{Z}^m, \sqrt{\Delta_{\mathbf{v}}}, \mathbf{x}},$$

where the Gaussian parameter is defined as $\Delta_{\mathbf{v}} = \Phi_{\mathbf{v}} - \Theta_{\mathbf{v}}$, which depends on the channel's Gaussian center $\mathbf{v}$ and is independent of $\mathbf{x}$.

Hyb₄: Almost identical to Hyb₃, except that Emb returns

$$\tilde{\mathbf{x}} \leftarrow \$ \mathcal{D}_{\mathbb{Z}^m, \sqrt{\Phi_{\mathbf{v}}}, \mathbf{v}},$$

where the Gaussian center is now $\mathbf{v}$. Note that this is identical to the RHS distribution in Eq. (2).

That $\text{Hyb}_0 \approx_s \text{Hyb}_1 \approx_s \text{Hyb}_2$ and $\text{Hyb}_3 \approx_s \text{Hyb}_4$ follow from Lemmas 19, 20 and 22 respectively, and the statistical distance between Hyb_2 and Hyb_3 is upper-bounded by $\frac{\beta_0 + \beta_1}{2(1 - \beta_0)}$ by Lemma 21. These lemmas are given in Appendix B.5.

Repeating the above $2N$ times, for each $i \in [N]$ and for both $\tilde{\mathbf{h}}_i$ and $\tilde{\mathbf{t}}_i$, we arrive at the claimed bounded. $\square$

From Theorems 1, 2 and 4 and Lemma 9, we obtain a CCA-secure PKS based on coversource-switching, given by Theorem 5 below. Moreover, here the PKS always utilises the channel's capacity in that its embedding rate is constant (instead of decreasing) in the channel's entropy.

Theorem 5 (CCA PKS for 0-memoryless TGC). *Let $(\mathcal{C}, \tilde{\mathcal{C}})$ be a pair of TGC (Definition 11) satisfying the conditions stated in Theorem 4 with the real numbers $\beta_0, \beta_1 \leq \text{negl}(\lambda)$. Let $\Pi_{\text{GaussSamp}}$ be the PKS Π in Fig. 2 instantiated with:*

- a correct ECC,
- a secure PRF,
- a correct and IND-\$-CPA secure PKE with a message space of size super-polynomial in λ , and
- the $\text{NSE}_{\text{GaussSamp}}$ in Fig. 4.

Then, $\Pi_{\text{GaussSamp}}$ is a correct and $(\mathcal{C}, \tilde{\mathcal{C}})$ -IND-CCA-secure PKS in the random oracle model.

Rate. For $\Pi_{\text{GaussSamp}}$ in Theorem 5, the stegotext space is of size approximately $\log |\mathcal{S}| \approx H(\mathcal{D}_{\mathbb{Z}^{m_H}, \tilde{s}}^N) \approx N \cdot m_H \cdot \log \tilde{s}$, where $H(\cdot)$ denotes the Shannon entropy, $\tilde{s}$ is the average Gaussian width of the channel $\tilde{\mathcal{C}}$, and the second approximation follows from elementary entropy bounds on Gaussians¹⁶. The message space is $\mathcal{M} = \mathcal{M}_H^N$ of size $\log |\mathcal{M}| = N \cdot n_H \log q$. By parameter choices, we have $m_H = n_H \lceil \log_\beta q \rceil$, and Theorem 4 requires $\tilde{s} - s \approx H(\tilde{\mathcal{C}}) - H(\mathcal{C})$ to be at least $\eta_\epsilon(\Lambda_q^\perp(\mathbf{G}_{n,\beta}))$ for some $\epsilon \leq \text{negl}(\lambda)$, where $\eta_\epsilon(\Lambda_q^\perp(\mathbf{G}_{n,\beta})) \leq \sqrt{1 + \beta^2} \cdot \omega(\sqrt{\log \lambda})$ by Lemmas 1 and 3. This implies $\tilde{s} \gtrsim s + \beta \cdot \omega(\sqrt{\log \lambda})$. By picking β appropriately such that the $\gtrsim$ is close to equality, overall, $\Pi_{\text{GaussSamp}}$ achieves a rate of

$$\begin{aligned} \log |\mathcal{M}| / \log |\mathcal{S}| &\approx (n_H \log q) / (n_H \log_\beta q \cdot \log \tilde{s}) \\ &= \log \beta / \log \tilde{s} \\ &\approx \log \beta / \log(s + \beta \cdot \omega(\sqrt{\log \lambda})) \approx 1 / \omega(\log \log \lambda), \end{aligned}$$

which is constant in the entropy $H(\tilde{\mathcal{C}})$ of $\tilde{\mathcal{C}}$.

6 Implementation for photo channels

To demonstrate practicality of our PKS for TGC based on Gaussian preimage sampling, we have implemented the PKS Π in Section 5.2 for RAW image photos.

6.1 Variant to handle real-life photo channels

To handle practical concerns such as uncertainty in image quality, we instantiate Π with a special choice of PKE and a variant of $\text{NSE}_{\text{GaussSamp}}$, call NSE_{Imp} , both given in Fig. 5. We sketch their rationales below.

The PKE in Fig. 5 calls an underlying PKE for N times, generating a set of N ciphertexts of the same ephemeral key ek , each under independent randomness. This takes care of e.g. settings where a photo has regional bad pixels, allowing the decoder to trial decrypt from the N possible ciphertexts, thus increasing reliability of the PKS. It is immediate that PKE is correct and IND- $\mathcal{S}$ -CPA secure if $\overline{\text{PKE}}$ is. NSE_{Imp} is similar to $\text{NSE}_{\text{GaussSamp}}$ but with multiple refinements:

- Two predicates $\mathcal{P}_H, \mathcal{P}_T$ are introduced which, given a head sample $\mathbf{h}_i$ resp. tail sample $\mathbf{t}_i$ from $\mathcal{C}$, determines the suitability to steg-embed on such sample.
- If $\mathcal{P}_H = 1$, i.e. “suitable”, EmbH calls Emb to generate $\text{head}_i = \tilde{\mathbf{h}}_i$ as expected; If $\mathcal{P}_H = 0$, i.e. “unsuitable”, EmbH instead calls a subroutine called SimEmbed , which simply samples a random Gaussian vector with parameters according to the channel description.
- For the tails, $\mathcal{P}_T$ is used by ChkT , where for each “unsuitable” sample $\mathbf{t}_i$, ChkT calls SimEmbed to sample a random Gaussian vector with parameters according to the channel description. The set $\mathcal{S}$ of “unsuitable” indices together with the random samples $(\tilde{\mathbf{t}}_i)_{i \in \mathcal{S}}$ are output as aux .
- EmbT uses aux to generate $\widetilde{\text{tails}}$, where it simply returns the random samples for the “unsuitable” indices, and it calls Emb to generate $\text{tail}_i = \tilde{\mathbf{t}}_i$ for the “suitable” indices.

The predicates $\mathcal{P}_H$ and $\mathcal{P}_T$ are to check if the noise difference between $\mathcal{C}_i$ and $\tilde{\mathcal{C}}_i$ is sufficient for performing Gaussian preimage-sampling. NSE_{Imp} has $(\mathcal{C}, \tilde{\mathcal{C}}, \epsilon)$ -indistinguishability and γ -tail-spreadness for the same ϵ, γ as $\text{NSE}_{\text{GaussSamp}}$, i.e. as given by Theorem 4. This follows from noting that SimEmbed is by design identical to the distribution in Hyb_2 in the proof of Theorem 4, i.e. the exact same proof applies. NSE_{Imp} has δ - and τ -correct head and tail embedding respectively for some δ, τ determined by the channel, i.e. actual photo quality. We instantiate ECC with one that is δ_{ECC} -error-correcting for some $\delta_{\text{ECC}} \geq \delta$ obtained empirically, which allows to correct the δ portion of errors in the embedded heads. We pick N sufficiently large so that empirically the failure probability of recovering ek from the embedded tails, which is $\leq \tau^N$, is sufficiently small.

¹⁶ We give related references in Appendix C.

PKE.Enc(pk, ek; r₁)	PKE.Dec(sk, ctxt = c)
parse $(r_{1,i})_{i \in [N]} \leftarrow r_1$	parse $(c_i)_{i \in [N]} \leftarrow c$
for $i \in [N]$ do	for $i \in [N]$ do
$c_i \leftarrow \text{PKE.Enc}(\text{pk}, \text{ek}; r_i) \quad \parallel c_i \in \mathbb{Z}_q^{n_T}$	$\text{ek} \leftarrow \text{PKE.Dec}(\text{sk}, c_i)$
$c := (c_i)_{i \in [N]}$	if $\text{ek} \neq \perp$ then return ek
return $\text{ctxt} := c$	return $\perp$
EmbH $^{\tilde{c} \geq i}(i, \mathbf{h}_i \in \mathbb{Z}^{m_H}, \mathbf{f}_i \in \mathbb{Z}_q^{n_H})$	EmbT $(i, \mathbf{t}_i \in \mathbb{Z}^{m_T}, \mathbf{c}_i \in \mathbb{Z}_q^{n_T}, \text{aux}; r)$
if $\mathcal{P}_H(\mathbf{h}_i) = 0$ then	parse $(S, (\tilde{\mathbf{t}}_i)_{i \in S}) \leftarrow \text{aux}$
return $\tilde{\mathbf{h}}_i \leftarrow \text{SimEmbed}(\mathbf{h}_i, m_H)$	if $i \in S$ then return $\tilde{\mathbf{t}}_i$
else return $\tilde{\mathbf{h}}_i \leftarrow \text{Emb}(\mathbf{h}_i, \mathbf{f}_i, n_H, \beta)$	else return $\tilde{\mathbf{t}}_i \leftarrow \text{Emb}(\mathbf{t}_i, \mathbf{c}_i, n_T, \beta; r)$
SimEmbed $(\mathbf{x}, m)$	ChkT $(\text{tails} = (\mathbf{t}_i)_{i \in [N]})$
$\parallel$ Gaussian param. according to $\mathbf{x}$:	$S := \emptyset \quad \parallel$ initialise empty set
$\Delta_{\mathbf{x}} := \Phi_{\mathbf{x}} - \Theta_{\mathbf{x}}$	for $i \in [N]$ do
return $\tilde{\mathbf{x}} \leftarrow \mathcal{D}_{\mathbb{Z}^m, \sqrt{\Delta_{\mathbf{x}}}, \mathbf{x}}$	if $0 = \mathcal{P}_T(\mathbf{t}_i)$ then
	$\tilde{\mathbf{t}}_i \leftarrow \text{SimEmbed}(\mathbf{t}_i, m_T); S := S \cup \{i\}$
	return $\text{aux} := (S, (\tilde{\mathbf{t}}_i)_{i \in S})$

Fig. 5. Implemented PKE and NSE_{imp} , for TGC being photo channels, with repetition of PKE encryption and simulated noise difference. $\mathcal{P}_H, \mathcal{P}_T$ are predicates. The algorithms Emb , ExtH , ExtT and Ext are identical to those in Fig. 4.

6.2 Experiment results

We have implemented the scheme described in Section 6.1 in around 1200 lines of Python code. The source code and the experiment results are publicly available as `ns-gembed`¹⁷.

The channel $\mathcal{C}$ is chosen to be ISO100 photos with $6000 \times 4000 = 24$ MP taken from a Sony A6400 camera. We consider three different choices of the channel $\tilde{\mathcal{C}}$: ISO200, 400 and 800 photos of the same camera saved in the digital negative (DNG) raw image format.¹⁸ The specification of DNG format is open and independent of cameras, thus it can support photos taken by all types of digital cameras. Further, it is a lossless raw format so that all integral pixel values within the range are valid pixel values. The noise profiles of Sony A6400 at these ISO levels, i.e. the Gaussian parameters Θ, Φ, Δ , are taken from [WYYL24].

We instantiate PRF with AES-256 in counter mode and ECC with the Reed-Solomon code over $\mathbb{F}_{256}$. We instantiate `SampPre` with that of [Pei10], ported from the C++ implementation in OpenFHE [BAB⁺22]. For PKE, we adapt the open-source implementation of ML-KEM-768, a.k.a. Kyber, from the `kyber-py` package¹⁹ to enable outputting multiple ciphertexts (c.f. Fig. 5). In our implementation, `aux` includes extra metadata to provide information for decoding, e.g. the base β and modulus q of the gadget matrix, and the error-correcting capacity δ_{ECC} of ECC. To optimise decoding runtime, we perform the re-encryption check (Step 14 of H.Dec in Fig. 2) after decoding $(\text{tail}, \text{aux})$ without decoding `msg`, where the former consumes only minimal space and time. This avoids redundant operations when some of the N trial PKE decryption fails to recover `ek`.

The best embedding rate is obtained by a grid search on the gadget matrix parameters β and $\lceil \log_{\beta} q \rceil$. Each pair of $(\beta, \lceil \log_{\beta} q \rceil)$ induces a different requirement on the Gaussian parameters Θ, Φ, Δ . The error-correcting

¹⁷ <https://github.com/gaussianstego/ns-gembed>

¹⁸ The camera produces images in Sony cRAW which is a lossy compressed RAW format, where some integral pixel values are out of the range of the decompression function. This means stego-embedding messages naively in this format will lead to trivial distinguishing attack – check if some pixel values lie outside of the range after decompression. We instead produce stego-embedded images in the DNG format and leave investigating embedding in the Sony cRAW format to future works.

¹⁹ <https://pypi.org/project/kyber-py/>

capacity δ_{ECC} of ECC is set dynamically based on the photo quality. Overall, the embedding rate is a function in $\beta, q, \delta_{\text{ECC}}$ where a larger β means embedding more bits per pixel but also higher failure probability at the same time, thus increasing β does not always imply a better embedding rate. We utilize just-in-time compilation for sequential operations on the photo, and PyMP for parallelisation. If the size of message to be embedded is below the maximum supported then the message is first padded before embedding.

We performed experiments on 12 everyday photos. On average, our stego-encoder takes 51.7 seconds, 35.7 seconds and 66.3 seconds to generate ISO-200, 400 and 800 stego photos respectively. The average ratios between the maximum message size and the uncompressed DNG file size are 7.9%, 19.3% and 24.7% for ISO-200, 400 and 800 respectively. These ratios correspond to the average message sizes 3.8MB, 9.4MB and 12.0MB, while the uncompressed DNG file size is 48.7MB. Note that the embedding rate would be higher if the message size is compared against the compressed DNG file size. The experimental results of individual photos are listed in Table 1.

		ISO-200			ISO-400			ISO-800		
Photo	DNG size	Msg. size	Runtime	Ratio	Msg. size	Runtime	Ratio	Msg. size	Runtime	Ratio
DSC00542	48674544	4277496	56.5s	8.8%	9392198	44.6s	19.3%	12048429	75.2s	24.8%
DSC00528	48674544	4062673	47.8s	8.3%	9390054	34.3s	19.3%	12046898	65.6s	24.7%
DSC00611	48674544	4207522	48.0s	8.6%	9392093	34.7s	19.3%	12048950	65.4s	24.8%
DSC00527	48674544	5997528	61.5s	12.3%	9414551	35.1s	19.3%	12070972	65.7s	24.8%
DSC00601	48674544	2502782	59.4s	5.1%	9395130	34.7s	19.3%	12053580	65.5s	24.8%
DSC00523	48674544	4249342	48.4s	8.7%	9396762	35.4s	19.3%	12056213	65.9s	24.8%
DSC00572	48674544	2014588	47.1s	4.1%	9387290	34.9s	19.3%	12041881	64.5s	24.7%
DSC00377	48674544	3468516	47.9s	7.1%	9371710	34.8s	19.3%	12024575	65.7s	24.7%
DSC00508	48674544	3823864	47.7s	7.9%	9381289	34.9s	19.3%	12033571	65.2s	24.7%
DSC00586	48674544	4431851	47.5s	9.1%	9402118	35.2s	19.3%	12059737	65.8s	24.8%
DSC00506	48674544	2771570	60.0s	5.7%	9384977	35.0s	19.3%	12039905	65.8s	24.7%
DSC00503	48674544	4303360	48.9s	8.8%	9371224	35.2s	19.3%	12027119	65.7s	24.7%

Table 1. The experimental results of individual photo. Sizes are in bytes.

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A Extended preliminaries

Lemma 10 ([GPV08]). *Fix any $\mathbf{A} \in \mathbb{Z}_q^{n \times m}$ satisfying $\mathbf{A} \cdot \mathbb{Z}_q^m = \mathbb{Z}_q^n$. Let $\epsilon \in (0, 0.5)$ and $\chi \geq \eta_\epsilon(\Lambda_q^\perp(\mathbf{A}))$. The statistical distance between the following is at most 2ϵ :*

$$\mathcal{D}_{\mathbb{Z}^m, \chi} \quad \text{and} \quad \left\{ \mathbf{x} \mid \mathbf{y} \leftarrow \mathbb{S}_{\mathbb{Z}_q^n}; \mathbf{x} \leftarrow \mathbb{S}_{\mathcal{D}_{\Lambda_q^\perp(\mathbf{A}), \chi}} \right\}.$$

Lemma 11 ([Pei10, Thm. 3.1]). *Let $\Sigma_1, \Sigma_2 > \mathbf{0}$ be positive definite matrices, with $\Sigma^* = \Sigma_1 + \Sigma_2 > \mathbf{0}$ and $\Sigma_3^{-1} = \Sigma_1^{-1} + \Sigma_2^{-1} > \mathbf{0}$. Let Λ_1, Λ_2 be m -dimensional lattices such that $\sqrt{\Sigma_2} > \eta_\epsilon(\Lambda_2)$ and $\sqrt{\Sigma_3} > \eta_\epsilon(\Lambda_1)$ for some positive $\epsilon \leq 1/2$, and let $\mathbf{c}_1, \mathbf{c}_2 \in \mathbb{R}^m$ be arbitrary. The statistical distance between the following is at most 8ϵ :*

$$\left\{ \mathbf{x}_2 \mid \mathbf{x}_2 \leftarrow \mathbb{S}_{\Lambda_2 + \mathbf{c}_2, \sqrt{\Sigma^*}} \right\} \quad \text{and} \quad \left\{ \mathbf{x}_2 \mid \begin{array}{l} \mathbf{x}_1 \leftarrow \mathbb{S}_{\Lambda_1 + \mathbf{c}_1, \sqrt{\Sigma_1}} \\ \mathbf{x}_2 \leftarrow \mathbb{S}_{\Lambda_2 + \mathbf{c}_2, \sqrt{\Sigma_2}, \mathbf{x}_1} \end{array} \right\}.$$

The following is a generalisation of Lemma 11.

Lemma 12. *Let $\Sigma_1, \Sigma^* \in \mathbb{R}^{m \times m}$ be positive definite matrices. Let $\mathbf{c} \in \mathbb{R}^m$ be arbitrary. For any $\mathbf{x} \in \mathbb{Z}^m$, let $\Sigma_{\mathbf{x}}^* \in \mathbb{R}^{m \times m}$ be positive definite matrix such that, for all $\mathbf{z} \in \mathbb{Z}^m$, it holds $\rho_{\sqrt{\Sigma_{\mathbf{x}}^*}}(\mathbf{z}) \in [1 - \beta_0, 1 + \beta_1] \cdot \rho_{\sqrt{\Sigma^*}}(\mathbf{z})$ for some $\beta_0, \beta_1 \in \mathbb{R}_{\geq 0}$. Then, the statistical distance between the distributions*

$$\left\{ \mathbf{x}_2 \mid \begin{array}{l} \mathbf{x}_1 \leftarrow \mathbb{S}_{\mathbb{Z}^m, \sqrt{\Sigma_1}, \mathbf{c}} \\ \mathbf{x}_2 \leftarrow \mathbb{S}_{\mathbb{Z}^m, \sqrt{\Sigma_{\mathbf{x}_1}^*}, \mathbf{x}_1} \end{array} \right\} \quad \text{and} \quad \left\{ \mathbf{x}_2 \mid \begin{array}{l} \mathbf{x}_1 \leftarrow \mathbb{S}_{\mathbb{Z}^m, \sqrt{\Sigma_1}, \mathbf{c}} \\ \mathbf{x}_2 \leftarrow \mathbb{S}_{\mathbb{Z}^m, \sqrt{\Sigma^*}, \mathbf{x}_1} \end{array} \right\}$$

is upper-bounded by $\frac{\beta_0 + \beta_1}{2(1 - \beta_0)}$.

Proof. Fix arbitrary $\bar{\mathbf{x}}_2 \in \mathbb{Z}^m$. By definition of the Gaussian distribution, in the RHS distribution

$$\begin{aligned} \Pr[\mathbf{x}_2 = \bar{\mathbf{x}}_2] &= \sum_{\bar{\mathbf{x}}_1 \in \mathbb{Z}^m} \Pr[\mathbf{x}_2 = \bar{\mathbf{x}}_2 \wedge \mathbf{x}_1 = \bar{\mathbf{x}}_1] \\ &= \sum_{\bar{\mathbf{x}}_1 \in \mathbb{Z}^m} \mathcal{D}_{\mathbb{Z}^m, \sqrt{\Sigma^*}}(\bar{\mathbf{x}}_2 - \bar{\mathbf{x}}_1) \cdot \mathcal{D}_{\mathbb{Z}^m, \sqrt{\Sigma_1}, \mathbf{c}}(\bar{\mathbf{x}}_1) \\ &= \sum_{\bar{\mathbf{x}}_1 \in \mathbb{Z}^m} \frac{\rho_{\sqrt{\Sigma^*}}(\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)}{\rho_{\sqrt{\Sigma^*}}(\mathbb{Z}^m)} \cdot \frac{\rho_{\sqrt{\Sigma_1}, \mathbf{c}}(\bar{\mathbf{x}}_1)}{\rho_{\sqrt{\Sigma_1}, \mathbf{c}}(\mathbb{Z}^m)}. \end{aligned}$$

Similarly, in the LHS distribution

$$\begin{aligned} \Pr[\mathbf{x}_2 = \bar{\mathbf{x}}_2] &= \sum_{\bar{\mathbf{x}}_1 \in \mathbb{Z}^m} \frac{\rho_{\sqrt{\Sigma_{\bar{\mathbf{x}}_1}^*}}(\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)}{\rho_{\sqrt{\Sigma_{\bar{\mathbf{x}}_1}^*}}(\mathbb{Z}^m)} \cdot \frac{\rho_{\sqrt{\Sigma_1}, \mathbf{c}}(\bar{\mathbf{x}}_1)}{\rho_{\sqrt{\Sigma_1}, \mathbf{c}}(\mathbb{Z}^m)} \\ &\in \left[\frac{1 - \beta_0}{1 + \beta_1}, \frac{1 + \beta_1}{1 - \beta_0} \right] \cdot \sum_{\bar{\mathbf{x}}_1 \in \mathbb{Z}^m} \frac{\rho_{\sqrt{\Sigma^*}}(\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)}{\rho_{\sqrt{\Sigma^*}}(\mathbb{Z}^m)} \cdot \frac{\rho_{\sqrt{\Sigma_1}, \mathbf{c}}(\bar{\mathbf{x}}_1)}{\rho_{\sqrt{\Sigma_1}, \mathbf{c}}(\mathbb{Z}^m)}. \end{aligned}$$

where we use the guarantee that $\rho_{\sqrt{\Sigma_{\mathbf{x}}^*}}(\mathbf{z}) \in [1 - \beta_0, 1 + \beta_1] \cdot \rho_{\sqrt{\Sigma^*}}(\mathbf{z})$ for any $\mathbf{z} \in \mathbb{Z}^m$, which also implies $\rho_{\sqrt{\Sigma_{\mathbf{x}}^*}}(\mathbb{Z}^m) \in [1 - \beta_0, 1 + \beta_1] \cdot \rho_{\sqrt{\Sigma^*}}(\mathbb{Z}^m)$. The concerned statistical distance is thus upper-bounded by

$$\frac{1}{2} \max \left(\left| 1 - \frac{1 - \beta_0}{1 + \beta_1} \right|, \left| 1 - \frac{1 + \beta_1}{1 - \beta_0} \right| \right) = \frac{\beta_0 + \beta_1}{2(1 - \beta_0)}.$$

□

A.1 Standard cryptographic primitives

Definition 12 (Pseudorandom Function (PRF)). A class $\mathcal{F}$ of functions $\mathcal{K} \times X \rightarrow Y$ is said to be pseudorandom functions (PRFs), if for all PPT $\mathcal{A}$ it holds that

$$\left| \Pr \left[1 \leftarrow \mathcal{A}^{\text{PRF}(\text{sk}, \cdot)} \mid \text{PRF} \leftarrow \mathcal{F}; \text{sk} \leftarrow \mathcal{K} \right] - \Pr \left[1 \leftarrow \mathcal{A}^{\text{O}(\cdot)} \right] \right| \leq \text{negl}(\lambda),$$

where the oracle $\text{O}(\cdot)$ outputs a uniformly random sample from Y on each new query, and it outputs the same answer for a repeated query.

We say that a function PRF is a secure PRF, if PRF is a random sample from $\mathcal{F}$.

A public-key encryption scheme can be seen as a public-key steganography scheme with a uniformly random channel.

Definition 13 (PKE). A public-key encryption (PKE) scheme $\text{PKE} = (\text{KGen}, \text{Enc}, \text{Dec})$ is a special case of a PKS (Definition 5), where the encoding algorithm Enc has no oracle access and does not input any history hist . The Enc and Dec algorithms are called encryption and decryption respectively, and the output of Enc is called a ciphertext ctxt .

A PKE is said to be correct if it is correct as a PKS (Definition 6), except that Enc has no oracle access.

A PKE scheme is said to be IND-\$-CCA (resp. IND-\$-CPA) secure, if it is $(\mathcal{C}, \tilde{\mathcal{C}})$ -IND-CCA (resp. $(\mathcal{C}, \tilde{\mathcal{C}})$ -IND-CPA) secure as a PKS (Definition 7), where $\mathcal{C} = \emptyset$ is empty and all distributions in $\tilde{\mathcal{C}}$ are the uniform distribution over the ciphertext space $\mathcal{S}$.

Definition 14 (Strongly-2-universal Hash Functions). A class $\mathcal{H}$ of functions $X \rightarrow Y$ is called strongly 2-universal if, for all distinct $x_1, x_2 \in X$ and all (not necessarily distinct) $y_1, y_2 \in Y$, exactly $|\mathcal{H}|/|Y|^2$ functions in $\mathcal{H}$ take x_1 to y_1 and x_2 to y_2 .

We say that a function H is a strongly-2-universal hash function, if H is a random sample from $\mathcal{H}$.

B Missing proofs for Section 5

B.1 Proof of Lemma 5

Proof. By design of $(\text{EncEK}, \text{DecEK})$, the stated event happens when both $\text{ExtT}(\text{EmbT}(i, \text{tail}_i, \text{msgt}, \text{ChkT}(\text{tails}))) = \text{msgt}$ and $\text{PKE.Dec}(\text{sk}, \text{PKE.Enc}(\text{pk}, \text{ek})) = \text{ek}$ occurs, which has probability at least $(1 - \text{negl}(\lambda))(1 - \text{negl}(\lambda)) = 1 - \text{negl}(\lambda)$ by the correctness of PKE and NSE. $\square$

B.2 Proof of Theorem 1

Proof. We inspect the code of II.Dec (Fig. 2).

Extract from $\widetilde{\text{tail}}_i$: By Lemma 5, II.Dec recovers ek from $\text{DecEK}(\text{sk}, (\widetilde{\text{tail}}_i)_{i \in [N]})$ with probability at least $1 - \text{negl}(\lambda)$ from its code line 5.

Extract from $\widetilde{\text{head}}_i$: Since G_1 and PRF are deterministic functions, whenever II.Dec recovers ek , it also recovers $(\text{sk}_{\text{PRF}}, r) \leftarrow \text{G}_1(\text{pk}, \text{ek})$ and $\text{PRF}(\text{sk}_{\text{PRF}}, i)$ with conditional probability 1. Then, by the head embedding correctness of NSE, II.Dec also recovers msgh_i and hence $\text{cw}_i = \text{msgh}_i \oplus \text{PRF}(\text{sk}_{\text{PRF}}, i)$ for all $i \in [N]$ with probability at least $1 - \text{negl}(\lambda)$ from its code lines 6–10.

Decode message: Conditioned on that II.Dec recovers all $\text{cw}_i, i \in [N]$, then by the correctness of ECC, II.Dec also recovers $(\text{msg}, \text{tails}, \text{aux})$ with conditional probability 1 from its code line 11.

Re-encryption check: Finally, II.Dec reruns G_2 and EncEK on $(\text{pk}, \text{ek}, \text{aux}, \widetilde{\text{heads}}, \text{tails})$. Both G_2 and EncEK are deterministic (since all randomness used in EncEK is deterministically derived from G_2). Hence, conditioned on that II.Dec successfully recovers ek and tails , the last check passes with probability 1.

Putting together, II.Dec correctly returns msg with probability at least $(1 - \text{negl}(\lambda))(1 - \text{negl}(\lambda)) = 1 - \text{negl}(\lambda)$. $\square$

$\text{DecO}(\text{stxt} = (\widetilde{\text{heads}}, \widetilde{\text{tails}})) \quad // \text{ Hyb}_0$
<pre> 1 : if stxt = stxt* then return $\perp$ 2 : ek $\leftarrow$ DecEK(sk*, $\widetilde{\text{tails}}$) 3 : (msg, tails, aux) $\leftarrow$ TrialDecrypt(pk*, ek, $\widetilde{\text{heads}}$) // run Π.Dec lines 6–11 with (pk*, ek, $\widetilde{\text{heads}}$) 4 : r $\leftarrow$ G₂(pk, ek, $\widetilde{\text{heads}}$, tails, aux) 5 : $\widetilde{\text{tails}}' \leftarrow$ EncEK(pk, ek, tails, aux; r) 6 : if $\widetilde{\text{tails}}' \neq \widetilde{\text{tails}}$ then return $\perp$ 7 : return msg </pre>
$\text{DecO}^\dagger(\text{stxt} = (\widetilde{\text{heads}}, \widetilde{\text{tails}})) \quad // \text{ Hyb}_1$
<pre> 1 : if stxt = stxt* then return $\perp$ 2 : if $\exists((\text{pk}, \text{ek}, \widetilde{\text{heads}}), r) \in \mathcal{L}_{G_2}$ 3 : (msg, tails, aux) $\leftarrow$ TrialDecrypt(pk, ek, $\widetilde{\text{heads}}$) // run Π.Dec lines 6–11 with (pk, ek, $\widetilde{\text{heads}}$) 4 : if (ek $\in \mathcal{K}$) $\wedge$ ($\widetilde{\text{tails}}$ = EncEK(pk, ek, tails, aux; r)) then 5 : return msg 6 : return $\perp$ </pre>

Fig. 6. Decryption oracle DecO and random oracle G₂ for Hyb₀ and Hyb₁ in proof of Theorem 2. pk*, sk* and ek* denote the public key, secret key and ephemeral key sampled when generating the challenge stegotext stxt* = (heads*, tails*), respectively.

B.3 Proofs for Theorem 2

Lemma 13. *If PKE is correct, and NSE has correct tail embedding and tail γ -spreadness where $\gamma \cdot N \geq \omega(\log \lambda)$, then $\text{Hyb}_0 \approx_s \text{Hyb}_1$.*

Proof. Throughout this proof, we will use the shorthand

$$(\text{msg}, \text{tails}, \text{aux}) \leftarrow \text{TrialDecrypt}(\text{pk}, \text{ek}, \widetilde{\text{heads}})$$

to refer to the subroutine of “running Π .Dec lines 6–11 on some $(\text{pk}, \text{ek}, \widetilde{\text{heads}})$ to obtain some $(\text{msg}, \text{tails}, \text{aux})$ ”.

In Hyb₁ we have swapped DecO to DecO[†] defined in Fig. 6, which does not make use of the secret key sk* anymore. Let $\mathcal{L}_{G_2}$ be the list of hash values recorded by the random oracle modelling G₂, i.e. $((\text{pk}, \text{ek}, \widetilde{\text{heads}}), r) \in \mathcal{L}_{G_2}$ means $(\text{pk}, \text{ek}, \widetilde{\text{heads}})$ has been queried and the answer is r .

Below, denote by the event $\mathcal{E} := “\mathcal{A}$ has queried DecO[†] on some $\text{stxt} = (\widetilde{\text{heads}}, \widetilde{\text{tails}}) \neq \text{stxt}^*$ where

$$\widetilde{\text{tails}} = \text{EncEK}(\text{pk}, \text{DecEK}(\text{sk}, \widetilde{\text{tails}}), \text{tails}, \text{aux}; G_2(\text{pk}, \text{ek}, \widetilde{\text{heads}}, \text{tails}, \text{aux})) \quad (3)$$

for some $\text{pk}, \text{ek}, \text{tails}, \text{aux}$, but there is no $((\text{pk}, \text{ek}, \widetilde{\text{heads}}), r) \in \mathcal{L}_{G_2}$ such that, for $(\text{msg}, \text{tails}, \text{aux}) \leftarrow \text{TrialDecrypt}(\text{pk}, \text{ek}, \widetilde{\text{heads}})$, it holds $\text{ek} \in \mathcal{K}$ and $\widetilde{\text{tails}} = \text{EncEK}(\text{pk}, \text{ek}, \text{tails}, \text{aux}; r)$ ”. Intuitively, $\mathcal{E}$ is the event where $\mathcal{A}$ has made a valid query to DecO[†] which it expects a decryption answer but DecO[†] returns $\perp$ (see Fig. 6).

First we show that, conditioned on $\neg \mathcal{E}$, the view of $\mathcal{A}$ in Hyb₀ and Hyb₁ are identical except if a correctness error has occurred in DecEK, i.e. $\text{DecEK}(\text{sk}, \text{EncEK}(\text{pk}, \text{ek}, \widetilde{\text{tails}}, \text{aux}; r)) \neq \text{ek}$, the latter happening with probability $\leq \text{negl}(\lambda)$ by Lemma 5. Fix any DecO[†] query $\text{stxt} = (\widetilde{\text{heads}}, \widetilde{\text{tails}}) \neq \text{stxt}^*$. Consider two cases:

- DecO in Hyb_0 returns $\text{msg} \neq \perp$. This means its lines 5–6 (Fig. 6) yields

$$\widetilde{\text{tails}}' = \text{EncEK}(\text{pk}, \text{ek}, \widetilde{\text{tails}}, \text{aux}; r) = \widetilde{\text{tails}}.$$

Then, because of $\neg \mathcal{E}$, $\text{DecO}^\dagger$ finds $((\text{pk}, \text{ek}, \widetilde{\text{heads}}), r) \in \mathcal{L}_{G_2}$ such that, for

$$(\text{msg}, \widetilde{\text{tails}}, \text{aux}) \leftarrow \text{TrialDecrypt}(\text{pk}, \text{ek}, \widetilde{\text{heads}}),$$

it holds that $\text{ek} \in \mathcal{K}$ and $\widetilde{\text{tails}} = \text{EncEK}(\text{pk}, \text{ek}, \widetilde{\text{tails}}, \text{aux}; r)$. Then by design of $\text{DecO}^\dagger$, it returns the same msg as in Hyb_0 since lines 6–11 of Dec is deterministic.

- DecO in Hyb_0 returns $\perp$. This means its lines 5–6 (Fig. 6) yields

$$\text{EncEK}(\text{pk}, \text{DecEK}(\text{sk}, \widetilde{\text{tails}}), \widetilde{\text{tails}}, \text{aux}; r) \neq \widetilde{\text{tails}}.$$

Then, unless there exists some $((\text{pk}', \text{ek}', \widetilde{\text{heads}}), r') \in \mathcal{L}_{G_2}$ such that, for

$$(\text{msg}', \widetilde{\text{tails}}', \text{aux}') \leftarrow \text{TrialDecrypt}(\text{pk}', \text{ek}', \widetilde{\text{heads}}'),$$

it happens that $\widetilde{\text{tails}} = \text{EncEK}(\text{pk}', \text{ek}', \widetilde{\text{tails}}', \text{aux}'; r')$, we have $\text{DecO}^\dagger$ also returns $\perp$. Conditioned on the correctness of $(\text{EncEK}, \text{DecEK})$, the said event is impossible, since it would imply

$$\text{DecEK}(\text{sk}, \widetilde{\text{tails}}) = \text{DecEK}(\text{sk}, \text{EncEK}(\text{pk}', \text{ek}', \widetilde{\text{tails}}', \text{aux}'; r')) = \text{ek}',$$

a contradiction to

$$\widetilde{\text{tails}} = \text{EncEK}(\text{pk}, \text{DecEK}(\text{sk}, \widetilde{\text{tails}}), \widetilde{\text{tails}}, \text{aux}; r) \neq \widetilde{\text{tails}}.$$

Next, we bound the probability that $\mathcal{E}$ happens. For any query $(\widetilde{\text{heads}}, \widetilde{\text{tails}}) \neq \text{stxt}^*$, denote $\text{ek} := \text{DecEK}(\text{sk}, \widetilde{\text{tails}})$, we argue that $(\text{pk}^*, \text{ek}^*, \widetilde{\text{heads}}^*) \neq (\text{pk}, \text{ek}, \widetilde{\text{heads}})$. We consider two cases:

- If $\widetilde{\text{tails}} = \widetilde{\text{tails}}^*$, then it must be that $\widetilde{\text{heads}} \neq \widetilde{\text{heads}}^*$; for otherwise the query equals stxt^* .
- If $\widetilde{\text{tails}} \neq \widetilde{\text{tails}}^*$, then it must be that $(\text{pk}^*, \text{ek}^*, \widetilde{\text{heads}}^*) \neq (\text{pk}, \text{ek}, \widetilde{\text{heads}})$; otherwise, we have $\widetilde{\text{tails}}^* = \text{EncEK}(\text{pk}, \text{ek}, \widetilde{\text{tails}}, \text{aux}; G_2(\text{pk}, \text{ek}, \widetilde{\text{heads}})) = \widetilde{\text{tails}}$ where the tails, aux are those from $(\text{msg}, \widetilde{\text{tails}}, \text{aux}) \leftarrow \text{TrialDecrypt}(\text{pk}, \text{ek}, \widetilde{\text{heads}})$, a contradiction.

Now since $(\text{pk}, \text{ek}, \widetilde{\text{heads}}) \neq (\text{pk}^*, \text{ek}^*, \widetilde{\text{heads}}^*)$ and G_2 is modelled as a random oracle, we have that $G_2(\text{pk}, \text{ek}, \widetilde{\text{heads}})$ is independent of $G_2(\text{pk}^*, \text{ek}^*, \widetilde{\text{heads}}^*)$. Together with Lemma 6, the probability that $\mathcal{E}$ happens is at most $q_{G_2} \cdot 2^{-\omega(\log \lambda)}$, where q_{G_2} denotes the number of queries made by $\mathcal{A}$ to G_2 .

Putting the above together, overall, the advantage of $\mathcal{A}$ in distinguishing Hyb_0 and Hyb_1 is upper-bounded by $\text{negl}(\lambda) + (1 - \text{negl}(\lambda)) \cdot q_{G_2} \cdot 2^{-\omega(\log \lambda)} \leq \text{negl}(\lambda)$. $\square$

Lemma 14. *If $|\mathcal{K}| \geq \lambda^{\omega(1)} \cdot \text{poly}(\lambda)$ and PKE is IND-CPA secure, then $\text{Hyb}_1 \approx_c \text{Hyb}_2$.*

Proof. The only difference between Hyb_1 and Hyb_2 is that the experiment aborts if the event $\mathcal{E} := “(\text{pk}^*, \text{ek}^*, \widetilde{\text{heads}})$ for some $\widetilde{\text{heads}}$ is queried on $G_2”$ happens, where we recall $\text{ek}^* \leftarrow \mathcal{K}$ is randomly sampled by the experiment. The advantage in distinguishing the stated two hybrids is thus identical to $\Pr[\mathcal{E}]$.

Below, we write $(\text{pk}, \text{ek}, \widetilde{\text{heads}}) \in G_2$ to denote the stated tuple has been queried by $\mathcal{A}$ on G_2 . We construct a PPT $\mathcal{B}$ against the IND-CPA security²⁰ of PKE, which succeeds if $\mathcal{E}$ happens. $\mathcal{B}$ proceeds as follows:

- Sample two random messages $\text{ek}_0 \leftarrow \mathcal{K}$, $\text{ek}_1 \leftarrow \mathcal{K}$.

²⁰ IND-CPA security of PKE, which says that no PPT $\mathcal{A}$ can distinguish a ciphertext encrypting either one of the two of its chosen messages, is implied by IND-CPA security of PKE.

- Pass (ek_0, ek_1) to the IND-CPA challenger, and receive from which the challenge $(pk^*, ctxt^*)$.
- Simulate the view of Hyb_1 or Hyb_2 to $\mathcal{A}$: It lets pk^* be the public key of Π and generates the challenge stegotext, where during the execution of $\Pi.Enc$ it sets $ctxt^*$ as the PKE ciphertext.
- Eventually receive some answer from $\mathcal{A}$.
- If $(pk^*, ek_{b'}, \widetilde{heads}) \in G_2$ but $(pk^*, ek_{1-b'}, \widetilde{heads}) \notin G_2$ then return b' . If both $(pk^*, ek_{b'}, \widetilde{heads}) \notin G_2$ and $(pk^*, ek_{1-b'}, \widetilde{heads}) \notin G_2$, then return a random bit $b' \leftarrow \{0, 1\}$.

Let the challenge $ctxt^*$ be encrypting ek_b , where $b \in \{0, 1\}$. Denote by $q_{G_2} \in \text{poly}(\lambda)$ the total number of queries on G_2 made by $\mathcal{A}$. The probability that $\mathcal{A}$ ever queries $(pk, ek_{1-b}, \widetilde{heads})$ on G_2 is at most $q_{G_2}/|\mathcal{K}|$, since ek_{1-b} is uniformly sampled from $\mathcal{K}$ and is information-theoretically hidden from $\mathcal{A}$. The rest is conditioned on that this does not happen.

If event $\mathcal{E}$ happens, this means $(pk^*, ek_b, \widetilde{heads}) \in G_2$, therefore $b' = b$. If event $\mathcal{E}$ does not happen, then both $(pk^*, ek_{b'}, \widetilde{heads}) \notin G_2$ and $(pk^*, ek_{1-b'}, \widetilde{heads}) \notin G_2$, hence the conditional probability of $b' = b$ is exactly $1/2$. Overall, we have

$$\begin{aligned} \text{Adv}_{\mathcal{B}, \text{PKE}}^{\text{ind-cpa}}(\lambda) + q_{G_2}/|\mathcal{K}| &\geq |\Pr[b' = b] - 1/2| \\ &= |\Pr[\mathcal{E}] + 1/2 \cdot \Pr[\neg \mathcal{E}] - 1/2| = 1/2 \cdot \Pr[\mathcal{E}], \end{aligned}$$

equivalently $\Pr[\mathcal{E}] \leq 2 \cdot (\text{Adv}_{\mathcal{B}, \text{PKE}}^{\text{ind-cpa}}(\lambda) + q_{G_2}/|\mathcal{K}|) \leq \text{negl}(\lambda)$, where the last inequality holds since $|\mathcal{K}| \geq \lambda^{\omega(1)} \cdot \text{poly}(\lambda)$. $\square$

Lemma 15. *If PKE is IND-\$-CPA secure, then $Hyb_2 \approx_c Hyb_3$.*

Proof. For any $\mathcal{A}$ distinguishing Hyb_2 and Hyb_3 , we construct a PPT $\mathcal{B}$ against the IND-\$-CPA security of PKE. $\mathcal{B}$ proceeds as follows:

- Sample a random $ek^* \leftarrow \mathcal{K}$. Pass ek^* to the IND-\$-CPA challenger, and receive from which the challenge $(pk, ctxt^*)$.
- Simulate the view of Hyb_2 or Hyb_3 to $\mathcal{A}$: It lets pk be the public key of Π and generates the challenge stegotext, where during the execution of $\Pi.Enc$ it sets $ctxt^*$ as the PKE ciphertext.
- Eventually receive b' from $\mathcal{A}$, and return b' .

We have that $\mathcal{B}$ perfectly simulates Hyb_2 if $ctxt^*$ encrypts ek^* , and it perfectly simulates Hyb_3 if $ctxt^*$ is a random sample from the PKE ciphertext space. The above follows since (1) sk^* is information-theoretically hidden from $\mathcal{A}$ from Hyb_1 onwards, and (2) Hyb_2 guarantees that $(pk^*, ek^*, \widetilde{heads})$ is never queried on G_2 by $\mathcal{A}$. Recall that r_1 in the output $r = (r_1, (r_{2,i})_{i \in [N]})$ of $G_2(pk^*, ek^*, \widetilde{heads})$ is used as the encryption randomness, which is thus uniformly random in the view of $\mathcal{A}$. $\square$

Lemma 16. *If $|\mathcal{K}| \geq \lambda^{\omega(1)} \cdot \text{poly}(\lambda)$, then $Hyb_3 \approx_s Hyb_4$.*

Proof. The only difference between Hyb_3 and Hyb_4 is that G_1 returns $\perp$ if being queried on (pk^*, ek^*) after the challenge stegotext is defined.

Recall that in Hyb_3 the PKE ciphertext is uniformly random and contains no information about ek^* . Hence, unless G_1 is queried on (pk^*, ek^*) , Hyb_3 and Hyb_4 are functionally identical. Since the view of $\mathcal{A}$ is independent of ek^* unless (pk^*, ek^*) is queried on G_1 , the probability that the latter happens is upper-bounded by $q_{G_1}/|\mathcal{K}| \leq \text{negl}(\lambda)$, where $q_{G_1} \in \text{poly}(\lambda)$ denotes the number of queries made by $\mathcal{A}$ to G_1 . $\square$

Lemma 17. *If PRF is secure, then $Hyb_4 \approx_c Hyb_5$.*

Proof. Observe that in Hyb_4 , no information about sk_{PRF}^* is provided to $\mathcal{A}$, since G_1 returns $\perp$ when being queried on the corresponding inputs (pk^*, ek^*) . Thus, sk_{PRF}^* is information-theoretically hidden from $\mathcal{A}$. The claim then follows from directly invoking PRF security. $\square$

Lemma 18. *If NSE has $(\mathcal{C}, \tilde{\mathcal{C}}, \epsilon)$ -indistinguishability for some $\epsilon \leq \text{negl}(\lambda)$, then $Hyb_5 \approx_s Hyb_6$.*

Proof. Note that in Hyb_5 , msgt_i is uniform over $\mathcal{M}_T^N$ and msg_i is uniform over $\mathcal{M}_H$ for each $i \in [N]$. The claim follows directly by the $(\mathcal{C}, \tilde{\mathcal{C}}, \epsilon)$ -indistinguishability of the said tuple of algorithms. $\square$

B.4 Proof of Lemma 8

Proof. That $\text{NSE}_{\text{RejSamp}}$ is PPT is clear: **EmbH** and **EmbT** each run λ iterations and each iteration is PPT, whereas **ExtH** and **ExtT** are deterministic.

Head embedding correctness: Correctness holds with probability 1 conditioned on **EmbH** does not return $\perp$, so it suffices to upper-bound the probability of the latter. Since H_1 is a strongly-2-universal hash and $H_\infty(\tilde{\mathcal{C}}_{\mathbb{H}})$ is sufficiently high, the entropy smoothing theorem [Lub96] implies that over the randomness of $(\text{head}_i, \text{tail}_i) \leftarrow \tilde{\mathcal{C}}_i$, $H_1(\text{head}_i)$ is statistically close in λ to a uniformly random sample over $\mathcal{M}_H$. Thus, for any fixed $\text{msg}_i \in \mathcal{M}_H$, we have $\Pr[H_1(\text{head}_i) \neq \text{msg}_i] \leq 1 - 1/|\mathcal{M}_H| + \text{negl}(\lambda) = 1 - \lambda^{-\gamma_H} + \text{negl}(\lambda)$, and the probability that **EmbH** returns $\perp$, i.e. the above event occurs for all λ iterations, is at most $(1 - \lambda^{-\gamma_H} + \text{negl}(\lambda))^\lambda \leq \text{negl}(\lambda)$.

Tail embedding correctness: Similarly, it suffices to upper-bound the probability of **EmbT** returning $\perp$. The definition of PSC implies that head_i and tail_i are independently distributed, and moreover $\{\text{tail}_i \mid \text{tail}_i \leftarrow \text{SampT}(i, \text{tail}_i)\} \equiv \{\text{tail}_i \mid (\text{head}_i, \text{tail}_i) \leftarrow \tilde{\mathcal{C}}_i\}$, hence it suffices to analyse the said probability for a modified **EmbT'** where **SampT** is replaced by $(\text{head}_i, \text{tail}_i) \leftarrow \tilde{\mathcal{C}}_i$. From here, the analysis is analogous to that for head embedding correctness, where we rely on that H_2 is a strongly-2-universal hash and $H_\infty(\tilde{\mathcal{C}}_{\mathbb{T}})$ is sufficiently high.

Indistinguishability: We need to show that Eq. (1) holds for some $\epsilon \leq \text{negl}(\lambda)$. Since head_i and tail_i are independently distributed for a sample $(\text{head}_i, \text{tail}_i) \leftarrow \tilde{\mathcal{C}}_i$ (implied by Definition 10), we can separately analyse the two. By Lemma 7, for any $(i, \text{head}_i, \text{msg}_i)$ resp. $(i, \text{tail}_i, \text{msg}_i, \text{aux})$, the statistical distance between

$$\begin{array}{ccc} \left\{ \text{head}_i \mid \text{head}_i \leftarrow \text{EmbH}^{\tilde{\mathcal{C}}_i}(i, \text{head}_i, \text{msg}_i) \right\} & \text{and} & \left\{ \text{head}_i \mid (\text{head}_i, \text{tail}_i) \leftarrow \tilde{\mathcal{C}}_i \right\}, \\ \text{resp.} \quad \left\{ \text{tail}_i \mid \text{tail}_i \leftarrow \text{EmbT}(i, \text{tail}_i, \text{msg}_i) \right\} & \text{and} & \left\{ \text{tail}_i \mid (\text{head}_i, \text{tail}_i) \leftarrow \tilde{\mathcal{C}}_i \right\}, \end{array}$$

are bounded by $2^{|\mathcal{M}_H| - H_\infty(\tilde{\mathcal{C}}_{\mathbb{H}})} + \alpha_H^\lambda \leq \text{negl}(\lambda)$ and $2^{|\mathcal{M}_T| - H_\infty(\tilde{\mathcal{C}}_{\mathbb{T}})} + \alpha_T^\lambda \leq \text{negl}(\lambda)$ respectively, where $\alpha_H, \alpha_T < 1$ are some constants. Combining the two, the statistical distance between the distributions in Eq. (1) is $\leq 2 \cdot N \cdot \text{negl}(\lambda) = \text{negl}(\lambda)$, where we recall $N \in \text{poly}(\lambda)$ is the length parameter of $\text{NSE}_{\text{RejSamp}}$.

Tail spreadness: The tail γ -spreadness property with $\gamma = H_\infty(\tilde{\mathcal{C}}_{\mathbb{T}})$ is implied by the definition of PSC. $\square$

B.5 Proofs for Theorem 4

Lemma 19. *If $\sqrt{\Delta_{\mathbf{x}}} \geq \eta_\epsilon(\Lambda_q^\perp(\mathbf{G}_{n,\beta}))$, then $\text{Hyb}_0 \approx_s \text{Hyb}_1$.*

Proof. Follows directly from Lemma 4 Item 2. $\square$

Lemma 20. *If $\sqrt{\Delta_{\mathbf{x}}} \geq \eta_\epsilon(\Lambda_q^\perp(\mathbf{G}_{n,\beta}))$, then $\text{Hyb}_1 \approx_s \text{Hyb}_2$.*

Proof. The claim follows from the Gaussian regularity lemma (Lemma 10) and the fact that $\mathbf{G}_{n,\beta}$ is primitive, i.e. $\mathbf{G} \cdot \mathbb{Z}_q^m = \mathbb{Z}_q^n$. To use Lemma 10, we rely on that in Hyb_1 , the input $\mathbf{y}$ to **Emb** is always uniform: It is either uniform $\mathbf{y} = \text{msg}_i \leftarrow \mathbb{Z}_q^{n_H}$, or uniform $\mathbf{y} = \text{msg}_i \leftarrow \mathbb{Z}_q^{n_T}$. $\square$

Lemma 21. *If $\rho_{\sqrt{\Delta_{\mathbf{x}}}}(\mathbf{z}) \in [1 - \beta_0, 1 + \beta_1] \cdot \rho_{\sqrt{\Delta_{\mathbf{v}}}}(\mathbf{z})$ for all $\mathbf{z} \in \mathbb{Z}^m$, then the statistical distance between Hyb_2 and Hyb_3 is upper-bounded by $\frac{\beta_0 + \beta_1}{2(1 - \beta_0)}$.*

Proof. The claim follows from Lemma 12, which says that the statistical distance between

$$\left\{ \tilde{\mathbf{x}} \mid \begin{array}{l} \mathbf{x} \leftarrow \mathcal{D}_{\mathbb{Z}^m, \sqrt{\Theta_{\mathbf{v}}}, \mathbf{v}} \\ \tilde{\mathbf{x}} \leftarrow \mathcal{D}_{\mathbb{Z}^m, \sqrt{\Delta_{\mathbf{x}}}, \mathbf{x}} \end{array} \right\} \quad \text{and} \quad \left\{ \tilde{\mathbf{x}} \mid \begin{array}{l} \mathbf{x} \leftarrow \mathcal{D}_{\mathbb{Z}^m, \sqrt{\Theta_{\mathbf{v}}}, \mathbf{v}} \\ \tilde{\mathbf{x}} \leftarrow \mathcal{D}_{\mathbb{Z}^m, \sqrt{\Delta_{\mathbf{v}}}, \mathbf{x}} \end{array} \right\}$$

is upper bounded by $\frac{\beta_0 + \beta_1}{2(1 - \beta_0)}$. $\square$

Lemma 22. *If $\sqrt{\Delta_{\mathbf{v}}} > \eta_{\epsilon}(\mathbb{Z}^m)$ and $\sqrt{\Theta_{\mathbf{v}}^{-1} + \Delta_{\mathbf{v}}^{-1}} > \eta_{\epsilon}(\mathbb{Z}^m)$, then $\text{Hyb}_3 \approx_s \text{Hyb}_4$.*

Proof. Follows directly from the Gaussian convolution lemma (Lemma 11), which says that

$$\left\{ \tilde{\mathbf{x}} \left| \begin{array}{l} \mathbf{x} \leftarrow \mathcal{D}_{\mathbb{Z}^m, \sqrt{\Theta_{\mathbf{v}}}, \mathbf{v}} \\ \tilde{\mathbf{x}} \leftarrow \mathcal{D}_{\mathbb{Z}^m, \sqrt{\Delta_{\mathbf{v}}}, \mathbf{x}} \end{array} \right. \right\} \approx_s \mathcal{D}_{\mathbb{Z}^m, \sqrt{\Phi_{\mathbf{v}}}, \mathbf{v}},$$

where we recall $\Phi_{\mathbf{v}} = \Theta_{\mathbf{v}} + \Delta_{\mathbf{v}}$ by definition of $\Delta_{\mathbf{v}}$. □

C Shannon entropy of $\mathcal{D}_{\mathbb{Z}, s, c}$

The definition of the Jacobi theta function is

$$\theta_3(z, q) := \sum_{n=-\infty}^{\infty} q^{n^2} e^{2niz} = 1 + 2 \sum_{n=1}^{\infty} q^{n^2} \cos(2nz),$$

where $i = \sqrt{-1}$, q is complex, and $|q| < 1$. It is continuous with respect to z when the imaginary part of $-(i \ln q)/\pi$ is positive. Thus, we can express the normalization constant $\rho_s(\mathbb{Z})$ by

$$\rho_{s,c}(\mathbb{Z}) := \sum_{n=-\infty}^{\infty} e^{-\pi(n-c)^2/s^2} = \sum_{n=-\infty}^{\infty} \left(e^{-\pi/s^2} \right)^{n^2} e^{2ni(-ic\pi/s^2)} e^{-\pi c^2/s^2} = e^{-\pi c^2/s^2} \theta_3(-ic\pi/s^2, e^{-\pi/s^2}).$$

For any complex z, x where $x \neq 0$, Jacobi's imaginary transformation gives

$$\theta_3(z, e^{-\pi x}) = x^{-1/2} e^{-z^2/\pi x} \theta_3(iz/x, e^{-\pi/x}).$$

Therefore, we have

$$\rho_{s,c}(\mathbb{Z}) = e^{-\pi c^2/s^2} \theta_3(-ic\pi/s^2, e^{-\pi/s^2}) = e^{-\pi c^2/s^2} \cdot s e^{\frac{c^2 \pi^2}{s^4} \frac{s^2}{\pi}} \theta_3(c\pi, e^{-\pi s^2}) = s \theta_3(c\pi, e^{-\pi s^2}).$$

Note that $\theta_3(z, q)$ has a period π of z , i.e., $\theta_3(z + \pi, q) = \theta_3(z, q)$. Thus we can set $c' = c \bmod 1$ and obtain

$$\rho_{s,c}(\mathbb{Z}) = s \theta_3(c\pi, e^{-\pi s^2}) = s \theta_3(c'\pi, e^{-\pi s^2}) = s + 2s \sum_{n=1}^{\infty} e^{-\pi n^2 s^2} \cos(2nc'\pi). \quad (4)$$

Lemma 23. *For $s > 0$, $\rho_{s,0.5}(\mathbb{Z}) \leq \rho_{s,c}(\mathbb{Z}) \leq \rho_{s,0}(\mathbb{Z})$ for all real c .*

Proof. We first prove the upper bound.

$$\begin{aligned} \rho_{s,c}(\mathbb{Z}) &= s + 2s \sum_{n=1}^{\infty} e^{-\pi n^2 s^2} \cos(2nc'\pi) \\ &\leq s + 2s \sum_{n=1}^{\infty} \left| e^{-\pi n^2 s^2} \cos(2nc'\pi) \right| \\ &\leq s + 2s \sum_{n=1}^{\infty} e^{-\pi n^2 s^2} \\ &= s + 2s \sum_{n=1}^{\infty} e^{-\pi n^2 s^2} \cos(2n(0)\pi) \\ &= \rho_{s,0}(\mathbb{Z}). \end{aligned}$$

For the lower bound, we consider the derivative of $\rho_{s,c}(\mathbb{Z})$ with respect to $c\pi$ and set it to 0 to find the minimizer $c\pi$.

$$\frac{d}{d(c\pi)} \rho_{s,c}(\mathbb{Z}) = s \frac{d}{d(c\pi)} \theta_3(c\pi, e^{-\pi s^2}) = 0.$$

The ratio of Jacobi theta function derivative to the function itself has the form²¹

$$\frac{\frac{d}{dz}\theta_3(z, q)}{\theta_3(z, q)} = -4 \sum_{n=1}^{\infty} \frac{q^{2n-1} \sin(2z)}{2q^{2n-1} \cos(2z) + q^{4n-2} + 1}.$$

Therefore, we have

$$s \frac{d}{d(c\pi)} \theta_3(c\pi, e^{-\pi s^2}) = 0 \Leftrightarrow \frac{\frac{d}{d(c\pi)} \theta_3(c\pi, e^{-\pi s^2})}{\theta_3(c\pi, e^{-\pi s^2})} = 0 \Leftrightarrow \sum_{n=1}^{\infty} \frac{e^{-\pi s^2(2n-1)} \sin(2\pi c)}{2e^{-\pi s^2(2n-1)} \cos(2\pi c) + (e^{-\pi s^2(2n-1)})^2 + 1} = 0.$$

Let $y(x) := 2x \cos(2\pi c) + x^2 + 1$ for real x . Note that the denominator inside the summation is $y(e^{-\pi s^2(2n-1)})$. The discriminant of $y(x)$ is $4 \cos^2(2\pi c) - 4 \leq 0$, thus $y(x) \geq 0$. On the other hand, the discriminant is 0 if and only if $\cos(2\pi c) = \pm 1$. In this case, $y(x)$ becomes $\pm 2x + x^2 + 1 = (x \pm 1)^2$, so the roots are $x = \mp 1$. When $x = e^{-\pi s^2(2n-1)} > 0$, the root -1 is rejected. Then, we have $e^{-\pi s^2} = 1$ and thus $s = 0$, which contradicts $s > 0$. Therefore, we can conclude that the denominator $2e^{-\pi s^2(2n-1)} \cos(2\pi c) + (e^{-\pi s^2(2n-1)})^2 + 1 > 0$.

For the numerator, we can see that

$$e^{-\pi s^2(2n-1)} \sin(2\pi c) \begin{cases} = 0 & \text{when } c \in \{0.5u : u \in \mathbb{Z}\}, \\ > 0 & \text{when } c \in \{v + u : 0 < v < 0.5, u \in \mathbb{Z}\}, \\ < 0 & \text{when } c \in \{v + u : 0.5 < v < 1, u \in \mathbb{Z}\}. \end{cases}$$

This means that all terms in the summation has the same sign. Therefore, we have

$$\sum_{n=1}^{\infty} \frac{e^{-\pi s^2(2n-1)} \sin(2\pi c)}{2e^{-\pi s^2(2n-1)} \cos(2\pi c) + (e^{-\pi s^2(2n-1)})^2 + 1} = 0 \Leftrightarrow c \in \{0.5u : u \in \mathbb{Z}\}.$$

Now we go back to the expression $\rho_{s,c}(\mathbb{Z}) = s\theta_3(c\pi, e^{-\pi s^2})$. The turning points are at $c \in \{0.5u : u \in \mathbb{Z}\}$. When have shown that $c = 0$ achieves the maximum. Due to periodicity of theta function, we know that $c \in \mathbb{Z}$ achieves the maximum. On the other hand, we have $-(i \ln e^{-\pi s^2})/\pi = is^2$ which has positive imaginary part, so $\theta_3(c\pi, e^{-\pi s^2})$ is continuous with respect to $c\pi$ and thus also continuous with respect to c . Due to continuity, we know that $c \in \{0.5 + u : u \in \mathbb{Z}\}$ achieves the minimum. This proves the desired lower bound. $\square$

Lemma 24. For $s > 0$, $s < \rho_{s,0}(\mathbb{Z}) < s + 2se^{-\pi s^2} + \operatorname{erfc}(\sqrt{\pi}s) < s + 2se^{-\pi s^2} + \frac{e^{-\pi s^2}}{\pi s}$, where $\operatorname{erfc}(z) := \frac{2}{\sqrt{\pi}} \int_z^{\infty} e^{-t^2} dt$ is the complementary error function.

Proof. As $se^{-\pi n^2 s^2} > 0$ for any $s > 0$ and any $n \in \mathbb{Z}$, Equation (4) gives $\rho_{s,0}(\mathbb{Z}) > s$.

Let $r(s) := s \sum_{n=1}^{\infty} e^{-\pi n^2 s^2} > 0$ be half the difference between $\rho_{s,0}(\mathbb{Z})$ and s . Note that $\frac{d}{dn} se^{-\pi n^2 s^2} = -2\pi n s^3 e^{-\pi n^2 s^2} < 0$ for any $s, n > 0$. This implies that $se^{-\pi n^2 s^2}$ is monotonically decreasing in n for $n > 0$. Therefore, we can upper bound $r(s)$ by

$$r(s) < se^{-\pi s^2} + \int_1^{\infty} se^{-\pi n^2 s^2} dn = se^{-\pi s^2} + \frac{s \operatorname{erfc}(\sqrt{\pi}|s|)}{2|s|}.$$

As $s > 0$, we have

$$r(s) < se^{-\pi s^2} + \frac{1}{2} \operatorname{erfc}(\sqrt{\pi}s). \quad (5)$$

On the other hand, for $z > 0$, an upper bound of $\operatorname{erfc}(z)$ is²²

$$\operatorname{erfc}(z) \leq \frac{2}{\sqrt{\pi}} \frac{e^{-z^2}}{z + \sqrt{z^2 + 4/\pi}}. \quad (6)$$

²¹ Equation (87) in <https://mathworld.wolfram.com/JacobiThetaFunctions.html>

²² Equation (13) in <https://mathworld.wolfram.com/Erfc.html>

Combining Equations (5) and (6), we have

$$r(s) < se^{-\pi s^2} + \frac{1}{\sqrt{\pi}} \frac{e^{-\pi s^2}}{\sqrt{\pi s} + \sqrt{\pi s^2 + 4/\pi}} < se^{-\pi s^2} + \frac{e^{-\pi s^2}}{2\pi s}.$$

Substituting it back to Equation (4), we obtain

$$\rho_{s,0}(\mathbb{Z}) = s + 2r(s) < s + 2se^{-\pi s^2} + \frac{e^{-\pi s^2}}{\pi s}. \quad \square$$

Lemma 25. For $s > 0$, $s > \rho_{s,0.5}(\mathbb{Z}) > s + \frac{1}{2}(\operatorname{erfc}(2\sqrt{\pi}s) - \operatorname{erfc}(\sqrt{\pi}s)) - 2se^{-\pi s^2} > s - \frac{e^{-\pi s^2}}{4\pi s} - 2se^{-\pi s^2}$.

Proof. From Equation (4), we know that

$$\rho_{s,0.5}(\mathbb{Z}) = s + 2s \sum_{n=1}^{\infty} e^{-\pi n^2 s^2} \cos(n\pi) = s + 2s \sum_{n=1}^{\infty} (-1)^n e^{-\pi n^2 s^2}.$$

The series $\sum_{n=1}^{\infty} (-1)^n e^{-\pi n^2 s^2}$ converges absolutely because

$$\sum_{n=1}^{\infty} \left| (-1)^n e^{-\pi n^2 s^2} \right| = \sum_{n=1}^{\infty} e^{-\pi n^2 s^2} = (\theta_3(0, e^{-\pi s^2}) - 1)/2.$$

Therefore, we can rearrange the terms into

$$\sum_{n=1}^{\infty} (-1)^n e^{-\pi n^2 s^2} = (-e^{-\pi s^2} + e^{-4\pi s^2}) + (-e^{-9\pi s^2} + e^{-16\pi s^2}) + \dots$$

As $e^{-\pi x^2 s^2}$ is monotonically decreasing in x for $x > 0$, every group on the right side is negative, implying that $\rho_{s,0.5}(\mathbb{Z}) < s$.

Now, we split the series into

$$\sum_{n=1}^{\infty} (-1)^n e^{-\pi n^2 s^2} = \sum_{n=1}^{\infty} e^{-\pi (2n)^2 s^2} - \sum_{n=1}^{\infty} e^{-\pi (2n-1)^2 s^2}.$$

Again, as $e^{-\pi x^2 s^2}$ is monotonically decreasing in x for $x > 0$, we have

$$\sum_{n=1}^{\infty} e^{-\pi (2n)^2 s^2} > \int_1^{\infty} e^{-\pi (2n)^2 s^2} dn = \frac{\operatorname{erfc}(2\sqrt{\pi}s)}{4s}$$

and

$$\sum_{n=1}^{\infty} e^{-\pi (2n-1)^2 s^2} < e^{-\pi s^2} + \int_1^{\infty} e^{-\pi (2n-1)^2 s^2} dn = e^{-\pi s^2} + \frac{\operatorname{erfc}(\sqrt{\pi}s)}{4s}.$$

Therefore, we have

$$\sum_{n=1}^{\infty} (-1)^n e^{-\pi n^2 s^2} > \frac{\operatorname{erfc}(2\sqrt{\pi}s)}{4s} - \frac{\operatorname{erfc}(\sqrt{\pi}s)}{4s} - e^{-\pi s^2}.$$

Thus

$$\rho_{s,0.5}(\mathbb{Z}) = s + 2s \sum_{n=1}^{\infty} (-1)^n e^{-\pi n^2 s^2} > s + \frac{1}{2}(\operatorname{erfc}(2\sqrt{\pi}s) - \operatorname{erfc}(\sqrt{\pi}s)) - 2se^{-\pi s^2}.$$

By dropping the term $\operatorname{erfc}(2\sqrt{\pi}s)/2$ and applying the upper bound in Equation (6), we have

$$\rho_{s,0.5}(\mathbb{Z}) > s - \frac{1}{2\sqrt{\pi}} \frac{e^{-\pi s^2}}{\sqrt{\pi s} + \sqrt{\pi s^2 + 4/\pi}} - 2se^{-\pi s^2} > s - \frac{e^{-\pi s^2}}{4\pi s} - 2se^{-\pi s^2}. \quad \square$$

By Lemmas 23 to 25, the upper bound and lower bound of $\rho_{s,c}(\mathbb{Z})$ approach s exponentially fast, suggesting that $\rho_{s,c}(\mathbb{Z})$ sticks close to s when s increases.

Lemma 26. $x^4 e^{-ax^2} \leq \frac{4}{a^2 e^2}$ for $a > 0$.

Proof. Let $y(x) := \frac{4}{a^2 e^2} - x^4 e^{-ax^2}$. We have $y'(x) = 2x^3 e^{-ax^2}(ax^2 - 2)$ and $y''(x) = -2x^2 e^{-ax^2}(2a^2 x^4 - 9ax^2 + 6)$. Set $y'(x) = 0$ to obtain turning points $x = 0, \pm\sqrt{2/a}$. As $y''(0) = 0$, $y'(x) > 0$ when $x \rightarrow 0^-$, and $y'(x) < 0$ when $x \rightarrow 0^+$, we know that $y(0)$ is a local maxima. On the other hand, $y''(\pm\sqrt{2/a}) = \frac{16}{ae^2} > 0$, we know that $y(\pm\sqrt{2/a}) = 0$ is a local minima. Therefore, we have $y(x) \geq 0$ for all x , which finishes the proof. $\square$

Theorem 6. Let $X \sim \mathcal{D}_{\mathbb{Z},s}$ and write $\rho_s(\mathbb{Z}) = s + t(s)$ where $s > 0$. The Shannon entropy $H(X)$ is

$$\begin{aligned} H(X) &= \log \rho_{s,c}(\mathbb{Z}) + \frac{\pi \log e}{s^2 \rho_{s,c}(\mathbb{Z})} \sum_{n=-\infty}^{\infty} (n - c')^2 e^{-\pi(n-c')^2/s^2} \\ &= \log \rho_{s,c}(\mathbb{Z}) + \frac{\pi e^{-\pi/s^2} \log e}{s^2} \frac{d \ln \rho_{s,c}(\mathbb{Z})}{d e^{-\pi/s^2}} \\ &= \log(s + t(s)) + \frac{s \log e}{2(s + t(s))} \left(1 + \frac{dt(s)}{ds} \right). \end{aligned}$$

Proof. We start with the definition:

$$\begin{aligned} H(X) &= - \sum_{n=-\infty}^{\infty} \frac{\rho_{s,c}(n)}{\rho_{s,c}(\mathbb{Z})} \log \frac{\rho_{s,c}(n)}{\rho_{s,c}(\mathbb{Z})} \\ &= \log \rho_{s,c}(\mathbb{Z}) + \frac{\pi \log e}{s^2 \rho_{s,c}(\mathbb{Z})} \sum_{n=-\infty}^{\infty} (n - c)^2 e^{-\pi(n-c)^2/s^2} \\ &= \log \rho_{s,c}(\mathbb{Z}) + \frac{\pi \log e}{s^2 \rho_{s,c}(\mathbb{Z})} \sum_{n=-\infty}^{\infty} (n - c')^2 e^{-\pi(n-c')^2/s^2} \\ &= \log \rho_{s,c}(\mathbb{Z}) + \frac{\pi e^{-\pi/s^2} \log e}{s^2 \rho_{s,c}(\mathbb{Z})} \sum_{n=-\infty}^{\infty} (n - c')^2 (e^{-\pi/s^2})^{(n-c')^2-1}, \end{aligned}$$

where $0 \leq c' := c \bmod 1 < 1$.

Let $f_n := (n - c')^2 (e^{-\pi/s^2})^{(n-c')^2-1} = e^{\pi/s^2} (n - c')^2 (e^{-\pi/s^2})^{(n-c')^2} \geq 0$. When $n \geq 2$ or $n \leq -1$, by Lemma 26, we have

$$f_n \leq e^{\pi/s^2} \frac{4}{(\pi/s^2)^2 e^2 (n - c')^2} \leq \begin{cases} \frac{4s^4 e^{\pi/s^2}}{\pi^2 e^2} \frac{1}{(n-1)^2} & \text{if } n \geq 2, \\ \frac{4s^4 e^{\pi/s^2}}{\pi^2 e^2} \frac{1}{n^2} & \text{if } n \leq -1. \end{cases}$$

Let

$$M_n := \begin{cases} \frac{4s^4 e^{\pi/s^2}}{\pi^2 e^2} \frac{1}{n^2} & \text{if } n \leq -1, \\ e^{\pi/s^2} (c')^2 (e^{-\pi/s^2})^{(c')^2} & \text{if } n = 0, \\ e^{\pi/s^2} (1 - c')^2 (e^{-\pi/s^2})^{(1-c')^2} & \text{if } n = 1, \\ \frac{4s^4 e^{\pi/s^2}}{\pi^2 e^2} \frac{1}{(n-1)^2} & \text{if } n \geq 2. \end{cases}$$

We can see that $0 \leq |f_n| \leq M_n$.

Recall that $\sum_{n=1}^{\infty} 1/n^2 = \pi^2/6$. We have

$$\sum_{n=-\infty}^{\infty} M_n = \sum_{n=-\infty}^{-1} \frac{4s^4 e^{\pi/s^2}}{\pi^2 e^2} \frac{1}{n^2} + M_0 + M_1 + \sum_{n=2}^{\infty} \frac{4s^4 e^{\pi/s^2}}{\pi^2 e^2} \frac{1}{(n-1)^2}$$

$$\begin{aligned}
&= \sum_{n=1}^{\infty} \frac{4s^4 e^{\pi/s^2}}{\pi^2 e^2} \frac{1}{n^2} + M_0 + M_1 + \sum_{n=1}^{\infty} \frac{4s^4 e^{\pi/s^2}}{\pi^2 e^2} \frac{1}{n^2} \\
&= e^{\pi/s^2} (c')^2 (e^{-\pi/s^2})^{(c')^2} + e^{\pi/s^2} (1-c')^2 (e^{-\pi/s^2})^{(1-c')^2} + \frac{4}{3} s^4 e^{\pi/s^2-2}.
\end{aligned}$$

By Weierstrass M-test, the series $\sum_{n=-\infty}^{\infty} f_n = \sum_{n=-\infty}^{\infty} \frac{d}{d e^{-\pi/s^2}} (e^{-\pi/s^2})^{(n-c')^2}$ converges absolutely and uniformly. So, we can swap the order of summation and differentiation to obtain

$$\begin{aligned}
H(X) &= \log \rho_{s,c}(\mathbb{Z}) + \frac{\pi e^{-\pi/s^2} \log e}{s^2 \rho_{s,c}(\mathbb{Z})} \frac{d}{d e^{-\pi/s^2}} \left(\sum_{n=-\infty}^{\infty} (e^{-\pi/s^2})^{(n-c')^2} \right) \\
&= \log \rho_{s,c}(\mathbb{Z}) + \frac{\pi e^{-\pi/s^2} \log e}{s^2 \rho_{s,c}(\mathbb{Z})} \frac{d \rho_{s,c}(\mathbb{Z})}{d e^{-\pi/s^2}} \\
&= \log \rho_{s,c}(\mathbb{Z}) + \frac{\pi \log e}{s^2 e^{\pi/s^2}} \frac{d \log \rho_{s,c}(\mathbb{Z})}{d e^{-\pi/s^2}}.
\end{aligned}$$

Substitute $\rho_{s,c}(\mathbb{Z}) = s + t(s)$, we have

$$H(X) = \log(s + t(s)) + \frac{\pi \log e}{s^2 e^{\pi/s^2}} \frac{d \log(s + t(s))}{d e^{-\pi/s^2}}.$$

Let $y(s) := e^{-\pi/s^2}$ where $s > 0$. Then, we have $s = \sqrt{\frac{\pi}{-\ln y(s)}}$. Thus,

$$\frac{ds}{d e^{-\pi/s^2}} = \frac{d}{dy} \sqrt{\frac{\pi}{-\ln y}} = \frac{\sqrt{\pi} \sqrt{-\ln y}}{2y \ln^2 y} = \frac{s^3 e^{\pi/s^2}}{2\pi}.$$

Then,

$$\begin{aligned}
\frac{d \log(s + t(s))}{d e^{-\pi/s^2}} &= \frac{1}{s + t(s)} \frac{d(s + t(s))}{d e^{-\pi/s^2}} \\
&= \frac{1}{s + t(s)} \left(\frac{ds}{d e^{-\pi/s^2}} + \frac{dt(s)}{ds} \frac{ds}{d e^{-\pi/s^2}} \right) \\
&= \frac{s^3 e^{\pi/s^2}}{2\pi(s + t(s))} \left(1 + \frac{dt(s)}{ds} \right).
\end{aligned}$$

Therefore, we have

$$H(X) = \log(s + t(s)) + \frac{s \log e}{2(s + t(s))} \left(1 + \frac{dt(s)}{ds} \right). \quad \square$$

Notice that $t(s)$ is very close to 0 when $s > \eta_\epsilon(\Lambda_q^\perp(\mathbf{G}))$, we can obtain a close approximation of Shannon entropy by setting $t(s) = 0$ and $t'(s) = 0$ in Theorem 6, i.e., we have $H(X) \approx \log s + \frac{1}{2} \log e$. To measure in unit bits, we take binary logarithm and conclude that

$$H_2(X) \approx \lg s + \frac{1}{2} \lg e = \lg s + \frac{1}{\ln 4} \approx \lg s + 0.72135 \text{ bits.}$$

When $0 \neq t(s) \approx 0$, it is not clear whether $t'(s) \approx 0$. Another way to approximate the entropy is to compute the integral

$$\int \frac{\pi}{s^2} x^2 e^{-\pi x^2/s^2} dx = \frac{s}{4} \operatorname{erf}(\sqrt{\pi} x/s) - \frac{1}{2} x e^{-\pi x^2/s^2} + \text{constant},$$

where $\operatorname{erf}(z) := \frac{2}{\sqrt{\pi}} \int_0^z e^{-t^2} dt$ is the error function. We have

$$\frac{\pi}{s^2} \sum_{n=-\infty}^{\infty} (n - c')^2 e^{-\pi(n-c')^2/s^2} \approx \int_{-\infty}^{\infty} \frac{\pi}{s^2} x^2 e^{-\pi x^2/s^2} dx = \frac{s}{2}.$$

We can numerically verify that the approximation is very close. This way, we have the same conclusion that

$$H_2(X) \approx \lg \rho_{s,c}(\mathbb{Z}) + \frac{s \lg e}{2\rho_{s,c}(\mathbb{Z})} \approx \lg s + \frac{1}{2} \lg e \approx \lg s + 0.72135 \text{ bits.}$$